

KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY



**A LOW-COST WIRELESS DIGITAL STETHOSCOPE
FOR CLINICAL USE IN GHANA**

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**PROJECT SUBMITTED TO THE DEPARTMENT OF COMPUTER ENGINEERING IN
PARTIAL FULFILLMENT OF THE REQUIREMENT FOR THE AWARD OF
BACHELOR OF SCIENCE IN BIOMEDICAL ENGINEERING**

DEDICATION

This project is dedicated to the diligent healthcare workers in Ghana who, by working tirelessly in resource-constrained environments, inspire innovations that close gaps in medical technology. It is also dedicated to our families and mentors, whose unwavering support and encouragement allowed us to start this journey.

ACKNOWLEDGEMENT

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ABSTRACT

Auscultation remains a vital diagnostic technique in clinical practice; however, conventional stethoscopes have significant drawbacks, including sensitivity to background noise, inability to record or share results, and failure to meet the demands of contemporary telemedicine. To overcome these obstacles, this project develops a wireless digital stethoscope that integrates with a smartphone app to record, share, and playback auscultation sounds in real-time. It is intended for use in healthcare settings with limited resources.

The system consists of an ESP32 microcontroller with integrated Bluetooth and a high-sensitivity electret microphone that records body sounds. This makes it possible for auscultation data to be wirelessly transmitted to a mobile application created with the Native Android framework. The app facilitates remote consultations and collaborative diagnosis by giving doctors an easy-to-use interface to view, store, and share patient recordings.

In comparison to traditional stethoscopes, the device's ability to preserve sound quality while drastically lowering ambient noise interference has been validated by preliminary testing conducted by qualified medical professionals. In PPE-intensive settings, the wireless design improves usability and removes the possibility of cross-contamination. In order to further enhance diagnostic capabilities, future developments will integrate AI-based sound analysis to automatically identify anomalies in heart and lung sounds.

This invention offers a portable, reasonably priced solution that fills the gap between conventional physical examination and contemporary telemedicine needs, marking a substantial advancement in digital auscultation technology. The gadget has a great potential to enhance healthcare delivery in both urban and rural areas by facilitating safe data sharing and remote diagnostics.

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1.0 CHAPTER 1: Introduction

1.1 Background of the study

Auscultation is the technique of using a stethoscope to listen for internal body sounds, primarily from the heart, lungs, and gastrointestinal tract. It involves using a stethoscope to listen to internal body sounds in order to identify abnormalities like breathing issues or heart murmurs. The method has progressed from placing the ears directly on the chest to using contemporary equipment that enhances sound collection and processing. The word "auscultation" comes from the Latin "auscultare," which means "to listen attentively." The phrase, which refers to pressing the ear against the chest wall, was first used by John Forbes. Jean-Baptiste Bouillaud used the stethoscope, which was later invented by René Laennec, to diagnose cardiac illness, but it was primarily used to detect lung sounds [1]. Depending on which body system is being studied, auscultation can take many different forms. By concentrating on heart sounds like normal S1 and S2 heartbeats, murmurs, gallops, and rubs, cardiac auscultation aids in the diagnosis of conditions like valve abnormalities and arrhythmias[2, 3]. Listening to breath sounds and identifying abnormalities like wheezing, crackles, or decreased airflow—all of which can indicate conditions like asthma, pneumonia, or chronic obstructive pulmonary disease (COPD)—is known as pulmonary auscultation [4, 5]. In order to diagnose gastrointestinal disorders like ileus or bowel obstruction, abdominal auscultation looks at bowel sounds, vascular bruits, and obstruction symptoms [6, 7]. Unusual blood flow sounds, such as bruits in the arteries, are detected by vascular auscultation and may indicate stenosis or aneurysms.

Despite its historical significance and ongoing application, the acoustic stethoscope has serious limitations that compromise its ability to diagnose conditions accurately. The stethoscope's chest piece makes a lot of noise with very low frequency movements, suggesting that the examiner must have a high level of hearing sensitivity [8]. It is impossible to share results with multiple practitioners at once because these stethoscopes only use direct auscultation, which only allows one doctor to hear the sound at a time.

Traditional stethoscope training is often subjective and relies on the instructor's interpretation rather than letting students hear and interpret the sounds themselves, which has an impact on medical education as well. The inability to record or amplify sounds makes training more

challenging because it deters students from repeating the sounds for a deeper understanding. Despite protocols that advise healthcare providers to wear face-covering personal protective equipment (PPE) and maintain social distancing, the COVID-19 pandemic presented a new challenge for the use of auscultation in clinical practice [9]. Healthcare personnel find it difficult to perform comprehensive examinations without taking off their protective gear because the physical barrier that comes with PPE typically reduces auscultation efficiency. Because many clinicians find traditional stethoscopes heavy and uncomfortable to carry around for extended periods of time, one of their biggest problems is their weight. Bacteria present in the stethoscope may cause healthcare-associated infections (HAIs). Approximately 80–100% of medical professionals' stethoscopes have been found to have two or more microorganisms on the ear pieces and/or the stethoscope diaphragm. Currently, disinfection is used to lower HAI, but low compliance frequently limits its effectiveness [10,11].

The rapid growth of telemedicine and digitization calls for the creation of technologically integrated medical devices that improve communication, accessibility, and efficiency in healthcare. Digital stethoscopes, which will offer notable advancements over conventional devices, are necessary to advance conventional auscultation. Physicians will be able to record, visualize, and share sounds for remote consultations and collaborative diagnosis thanks to these digital stethoscopes' sound amplification, noise reduction, and digital storage features [12, 13]. The goal of this project is to create a digital stethoscope that supports telemedicine, medical education, and infection control applications by incorporating noise filtering, real-time wireless transmission, and data storage features.

1.2 Problem Statement

In medical diagnostics, accurate auscultation is crucial because it enables the identification of cardiovascular and respiratory conditions in addition to other abdominal issues. The accuracy of diagnosis is impacted by auscultation, which mainly depends on the clarity of sound detected by stethoscopes. The effectiveness of conventional and electronic stethoscopes has been the subject of numerous studies, which have shown notable variations in sound amplification and clarity. Even though they are widely used, traditional stethoscopes often have trouble making clear sound, especially in noisy environments. For instance, studies have demonstrated that the reverse stethoscope technique hinders the comprehension of speech by patients with hearing impairments [14]. Research has indicated that while auscultation can be performed in many hospital emergency rooms, where background noise levels range from 60 to 70 dB, it is not feasible to detect heart murmurs and other clinical abnormalities [15, 16]. Because cardiac sounds are very low frequency and even small movements of the stethoscope chest piece create a lot of noise during cardiac auscultation, the examiner must have an acute sense of hearing. Given how quickly the heart beats, losing concentration could lead to the omission of important details and an incorrect diagnosis [8, 17]. Additionally, unlike digital stethoscopes, conventional stethoscopes can only be used by one operator at a time. This means that the sound data cannot be processed, exchanged, analyzed, or used in any other way. Because of this, it is not as helpful when demonstrating chest auscultation for educational purposes or when playing back previous audio readings for cross-referencing during the diagnosis process [18, 19]. The COVID-19 pandemic presented a new challenge for the use of auscultation in clinical practice, despite standards advising healthcare workers to wear face-covering personal protective equipment (PPE) and maintain social distancing. Since their protective gear covers their ears, medical personnel in quarantine wards in hospitals are unable to use conventional stethoscopes [20].

1.3 [Objectives of the study](#)

1.3.1 General Objective

To design an affordable wireless digital stethoscope that supports telemedicine, medical education, and patient record keeping by incorporating mobile connectivity, noise reduction, and amplification for improved auscultation.

1.3.2 Specific Objectives

- Design and develop an affordable Bluetooth-enabled wireless digital stethoscope that improves sound quality for mobile device playback by combining noise reduction and amplification.
- Develop a smartphone app that allows auscultation sounds to be played back, recorded, and shared in real time to facilitate digital patient documentation, telemedicine, and medical education.
- Utilize affordable resources while preserving signal quality to guarantee accessibility for medical personnel.

1.4 [Scope of the Study](#)

This project focuses on the design, development, and evaluation of a low-cost wireless digital stethoscope tailored for clinical use in Ghana. Study covers both software and hardware development, guaranteeing usability, affordability, and mobility in healthcare environments with limited resources. Heart sounds will be transmitted in real time via Bluetooth from the device to a mobile application, where algorithms for noise reduction and amplification will improve the signal quality for more audible auscultation. Furthermore, the application will facilitate telemedicine, medical education, and patient data documentation by supporting sound recording, storing, and sharing. The project's objective is to create a working prototype with affordable materials while preserving diagnostic precision and signal integrity.

1.5 Limitations and Exclusions

This project does not involve mass production, regulatory certification (such as FDA), or comprehensive clinical validation; instead, it focuses on creating a proof-of-concept prototype within a 6-month academic timeframe. Cost-effectiveness and functionality will be given top priority in the design, although exact costs may differ depending on Ghana's local component availability. Advanced cloud-based features or real-time remote consultations are currently outside the purview of the mobile application, even though it will support Bluetooth connectivity, noise filtering, and playback. Future stages of development will take these factors into account.

1.6 Significance of the Study

This project aligns with Sustainable Development Goal 3 (SDG 3) by improving access to affordable and efficient diagnostic tools in Ghana's healthcare system. The accuracy of auscultation, a crucial medical diagnosis skill, is impacted in many healthcare facilities, particularly in rural areas, by limited access to high-quality stethoscopes and training aids. By providing real-time sound transmission and playback on mobile devices via Bluetooth connectivity, the development of an affordable, portable, wireless digital stethoscope seeks to improve auscultation. This will enhance sound quality, lower the possibility of ear infections from sharing stethoscopes, and make it simple to record and share audio for remote diagnosis and consultation. In addition to its clinical uses, this stethoscope will be a useful teaching tool for medical professionals and students, enabling them to record, filter, and replay auscultation sounds for better training and education. By solving the problems with traditional stethoscopes, this invention improves medical education, healthcare delivery, and diagnostic precision in Ghana while increasing the effectiveness and accessibility of contemporary auscultation methods.

1.7 Organization of the Report

This report is systematically organized into six chapters, each contributing to a comprehensive documentation of the project's development and outcomes:

- Chapter 1 (Introduction): Lays the groundwork for the study by giving a summary of the project's background, problem statement, significance, goals, and scope.
- Chapter 2 (Literature Review): Identifies gaps in the literature and places the contributions of this project in context by reviewing the literature on stethoscope technology, wireless transmission systems, and digital auscultation.
- Chapter 3 (Methodology): the technical and procedural steps taken to develop the prototype are described, along with the hardware and software design, signal processing techniques, and system integration.
- Chapter 4 (Results): The results of the prototype's performance evaluation are presented, which also covers sound quality analysis, the efficiency of noise filtering, and the dependability of Bluetooth transmission.
- Chapter 5 (Discussion): Examines the practical implications of the digital stethoscope for clinical use and medical education in Ghana, analyzes the findings, and discusses the difficulties faced.
- Chapter 6 (Conclusion and Recommendations): summarizes the study, points out its limitations, and makes suggestions for future improvements, possible clinical validation, and additional research.

2.0 CHAPTER 2: Literature Review

2.1 Introduction

The development of stethoscope technology has significantly enhanced clinical diagnosis by enabling physicians to successfully assess cardiovascular and respiratory conditions. Notwithstanding their widespread use, traditional acoustic stethoscopes have drawbacks, such as insufficient sound amplification, background noise interference, and the inability to store or distribute auscultation data. The difficulties have been addressed by the creation of digital stethoscopes, which enhance auscultation accuracy by combining sophisticated signal processing, noise reduction, and wireless transmission.

An extensive overview of the latest developments in digital and wireless stethoscope research, technology, and advancements is provided in this chapter. The modernization of stethoscope capabilities is discussed, including the ideas of auscultation, the transition from analog to digital stethoscopes, and the role of mobile apps, Bluetooth connectivity, and noise reduction. The chapter also highlights the research gaps that this project aims to address and lists the most important drawbacks of the current models of digital stethoscopes. By discussing these elements, this chapter establishes the framework for developing a wireless, reasonably priced digital stethoscope that is appropriate for useful and efficient clinical use.

2.2 Overview of Conventional Stethoscopes

2.2.1 Auscultation

Auscultation is a key clinical skill used to assess heart and lung disorders by interpreting the sounds produced by the body. Despite advances in technology, the combination of contemporary and conventional auscultation techniques enhances diagnostic precision and educational results. It is crucial for both diagnosing and tracking diseases, especially those that impact the cardiovascular, gastrointestinal, and respiratory systems. Intestinal, lung, and heart sounds are the three primary sound categories evaluated during auscultation. Between 20 and 150 Hz, heart sounds, most notably S1 and S2, are produced. Additional heartbeats or murmurs are examples of abnormalities in these sounds that could indicate structural or functional heart disorders. The characteristics of lung sounds, such as wheezes, crackles, and rhonchi, can be used to diagnose asthma, pneumonia, and chronic obstructive pulmonary disease (COPD). Typically, these noises range from 100 Hz to

2,500 Hz. On the other hand, bowel noises, which range from 100 Hz to 500 Hz, aid in identifying irregularities in bowel movements. Comprehending the frequency ranges and clinical significance of these sounds enhances the precision of diagnosis and guides the appropriate medical interventions.

2.2.2 History and Evolution of the Stethoscope

The stethoscope is more closely associated with the idea of being a doctor than any other symbol. It is among the most widely used diagnostic aids by doctors and nurses. Prior to the invention of the stethoscope, auscultation was practiced by medical professionals who placed their ear on the patient's chest using a technique that became known as "immediate auscultation" (i.e., without the interposition of another item or mediator). Auscultation may therefore be useful in the diagnosis of lung conditions, but it was not very useful in the diagnosis of heart conditions. The invention of the stethoscope was revolutionary in the diagnosis and treatment of cardiac and pulmonary disorders, despite its apparent simplicity and logic. René Theophile Hyacinthe Laennec (1781–1826), a physician from Breton who studied under Corvisart and later worked as a physician at the renowned Saltpetre and Necker Hospital in Paris, invented the first stethoscope. It was a unitubular cylinder that was initially constructed in 1819 out of paper and pasteboard before being converted into a wooden tube. He gave it the name "LE CYCLINDER".

Laennec, who immediately recognized the stethoscope's potential as a scientific and therapeutic tool, skillfully illustrated how it also created new opportunities for early research into the pathophysiology of chest disorders. In 1816, French physician Laennec was notified of a young woman who was overweight and showing symptoms of heart disease. He felt that immediate auscultation was inappropriate due to the "age and sex of the patient." Then he remembered a well-known acoustic phenomenon: if the ear is placed on one end of a log, the tapping of a pin at that end can be clearly heard at the other end. Inspired by this observation, he formed a tight roll out of a paper notebook, placed one end against the patient's precordium, and placed the other end against his ear, and was "as surprised as satisfied" to find out how well he could hear the heart sounds [21].



He then transformed his idea into a hollow wooden monaural tube with a funnel at one end. Its three parts were the core shaft, the funnel-shaped part for listening to the lungs, and an extra plug piece with a brass tube that could be put into the funnel to listen to the heart. The stethoscope had a 3/8-inch central lumen and measured

12.6 inches in length by 1.5 inches in width. Thus, in 1816, the first stethoscope was invented. It

Figure 1 Laennec's stethoscope circa 1819

was easily separated into two sections, one of which screwed into the other, and featured a detachable chest piece at one end. His contemporaries wanted to call it "somascopes," "ethoscopes," or "Echophore" because it was used to examine more than just the chest. Laennec described the design of his stethoscope in detail in his 1819 Paris publication, "De L'auscultation médiate." In those days, wooden stethoscopes made to his specifications were sold in Piccadilly, London, for four shillings each. It was monaural and only affected one ear [22, 23]. In 1828, Pierre Adolphe Piorry proposed the diaphragm, a device that used a piece of ivory breast as a pleximeter. The shapes of the bell and chest piece were changed, as were the materials used to construct the instrument [24].

An early desire to create such an instrument led Londoner CJB Williams to use two bent lead pipes to create a binaural instrument in 1829. For the tube to advance, a flexible material had to be obtained. In 1855, the first binaural stethoscope was introduced by Dr. George Cammann. Cammann's stethoscope was similar to modern stethoscopes in that it had two tubes from the ear parts that extended to connect a single bell-shaped chest piece. Rubber, metal, and silk-covered metal wire springs were among the materials used to make the stethoscope tubes.

In the middle of the 1800s, Allison created the differential stethoscope. This equipment included two tubes that transmitted sound from the chest to the examiner's ears.

Unlike Cammann's stethoscope, the two tubes ended in two separate bells, which allowed the examiner to simultaneously auscultate two different areas of the chest. As one might expect, this was confusing and not helpful [25].

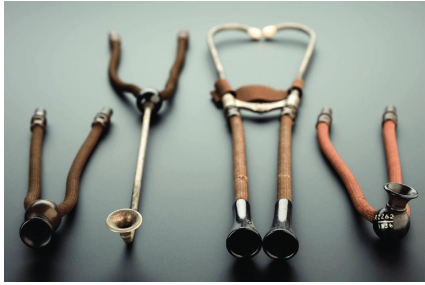


Figure 3 Allison's differential stethoscope,

The two bell chest pieces were used to

Auscultate two different spots on the chest



*Figure 2 George Cammann
Stethoscope*

The most creative and thoughtful design change was made in 1894 when Robert Bowles of Massachusetts developed the diaphragm-style chest component. The flat surface was easy to apply to the elbow and made good contact with the chest when taking blood pressure. It wasn't until much later that its superiority over the bell for high-pitched sounds was recognized [26]. In 1926, Howard Sprague of Boston made the final major modification by fusing a diaphragm and bell into a single chest piece. Even into the 1950s, many physicians still just used a bell or occasionally a diaphragm, but the Spaghe-Bowles stethoscope eventually became the standard tool. As Dr. Blafox points out, there have been a lot of changes over the years, and he has done a fantastic job of simplifying a complex problem. To avoid being biased, I suggest that he have used the 1955 design made by Aubrey Leatham at London's St. George's Hospital, which included an unusual bell with two parts: a large bell and 626 smaller bells. The Littmann stethoscope, which was essentially a redesign of the Sprague–Bowles in 1961, is currently the industry standard due to its portability and ease of use [27, 28].

In his book's conclusion, Dr. Blafox regrettably states that the advancement of modern imaging techniques has caused clinical examination to be overlooked. The basic skills of taking a history, observing, and examining are negated by the devices that take the place of the stethoscope [29, 30].

2.2.3 Parts of a stethoscope

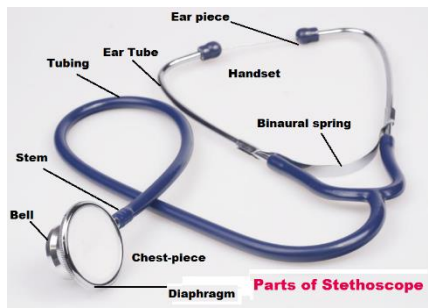


Figure 4 parts of a stethoscope

Source:

<https://www.medilogbiohealth.com/2021/03/parts-of-stethoscope.html>

Healthcare professionals use the acoustic stethoscope as a diagnostic tool to listen for internal body sounds, particularly those originating from the heart, lungs, and intestines. It transmits sound through hollow tubes rather than amplifying it electronically.

The main components and their functions are:

1. Chest piece (Head)

- **Function:** Captures sounds from the body.
- **Parts:**
 - **Diaphragm:** This is a flat, circular membrane used to detect high-frequency sounds from the body (e.g., heartbeats, breath sounds).
 - **Bell:** This is a smaller, concave component used for detecting low-frequency sounds (e.g., certain heart murmurs). It is attached to the diaphragm

2. Stem

- **Function:** It connects the chest piece to the tubing and allows switching between the diaphragm and bell.
- The diaphragm or the bell can be activated by a rotating mechanism in certain models.

3. Tubing

- **Function:** Transmits sound from the chest piece to the earpieces.
- Made of flexible rubber or PVC which insulates sound and reduces ambient noise interference.

4. Headset

- **Function:** Transmits sound from the tubing to the ears.
- It consists of :
 - **Binaural:** Metal tubes that connect the tubing to the earpieces.
 - **Spring mechanism:** Ensures tension to keep the headset snug in the ears.

5. Earpieces (Ear Tips)

- **Function:** Deliver sound into the user's ears.
- Made of soft rubber or silicone for comfort and acoustic sealing.

2.2.4 Limitations of traditional acoustic stethoscopes.

The acoustic stethoscope is a medical instrument, but it has some limitations that reduce its effectiveness when performing auscultation. Accurate diagnosis is hindered by these limitations, which stem from both technological and user-related issues. One major drawback of acoustic stethoscopes is their poor sound clarity, which is particularly problematic in noisy environments or when assessing patients with weak lung or heart sounds [19]. Auscultation is nearly impossible in circumstances where background noise levels may exceed 80 dB, such as during medical evacuations, and acoustic stethoscopes are ineffective in these circumstances [31]. Due to their inability to record or transmit auscultatory sounds, acoustic stethoscopes are not appropriate for remote diagnostics in telemedicine applications. This restriction makes it more difficult to remotely share and analyze patient data, which is necessary for telehealth services to be effective. Because they cannot record or replay sounds, students cannot assess and learn from cases outside of actual clinical encounters. Furthermore, the absence of simultaneous listening capabilities makes it impossible for teachers to verify what students hear during auscultation, which reduces the effectiveness of evaluation and feedback [32]. Acoustic stethoscopes are difficult to use effectively when full Personal Protective Equipment (PPE) is needed, such as gowns, gloves, and especially face shields or hoods. These barriers not only make it challenging to position the earpieces correctly, but they also compromise the seal for optimal sound transmission. This restriction was particularly evident during the COVID-19 pandemic, when infection control measures hindered the use of traditional stethoscopes [20].

2.2.5 Field Data Supporting Identified Limitations

A structured questionnaire was used to gather field data from about 100 clinicians, including medical students undergoing clinical rotations, in order to verify the usefulness of acoustic stethoscopes. Key issues like poor sound clarity, no digital recording, incompatibility with personal protective equipment, and unsuitability for telemedicine were highlighted by the analysis and graphic representation of the responses.

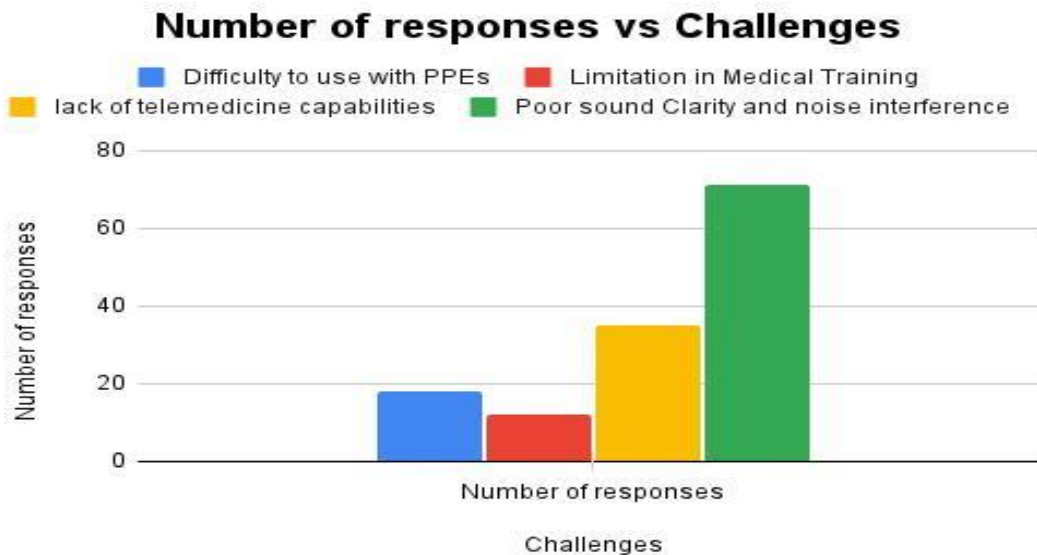


Figure 5 A graph of the number of responses against limitations

2.3 The Digital Stethoscope

Digital stethoscopes, which use modern technology to improve auscultation techniques, have greatly improved medical diagnosis. Real-time analysis and improved diagnostics are made possible by these devices, which transform acoustic noises into digital information. By transforming sound data into electrical impulses that can be amplified, stored, replayed, and—most importantly—transmitted for expert judgment, the digital stethoscope overcomes the drawbacks of a traditional stethoscope, making it incredibly useful in telemedicine[33]. Certain devices increase the efficiency of diagnosis by combining oximetry and stethoscope capabilities, which allows for the simultaneous monitoring of cardiac activity and oxygen saturation levels [34]. Artificial intelligence is used in digital stethoscopes to improve signal processing, which enables accurate sound analysis and the identification of irregularities in lung and heart sounds [35].

The hardware of a digital stethoscope is made up of a number of crucial parts that combine to capture and process bodily sounds like heartbeats. Acoustic impulses are detected from the chest by an electret condenser sensor or a microphone on the front end. Before being processed further, these signals are amplified by a preamplifier because they are often weak. Only the crucial frequency range of heart or lung sounds remains after the noise is eliminated using an active filter. After filtering, a primary amplifier may be used to boost the signal's volume for analysis or listening. After being processed by a microcontroller, the sound is then transmitted via Bluetooth or USB modules to a computer or smartphone. Every part of the system is connected on a circuit board inside the stethoscope casing, and it is powered by a tiny battery.

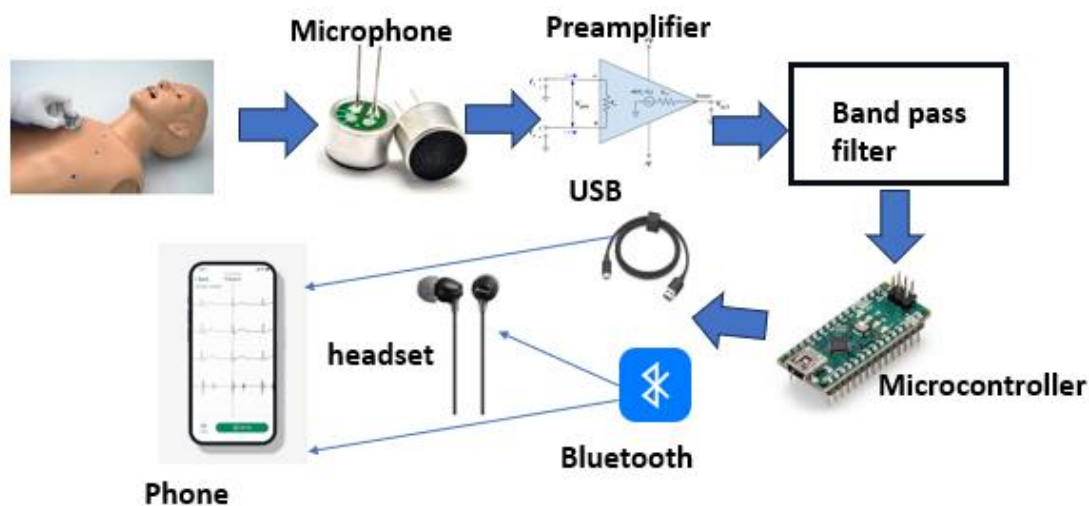


Figure 6 Basic operation of a digital stethoscope

The hardware of the digital stethoscope consists of a microphone (such as the ECM-10 or MAX9814) that records lung and heart sounds. A preamplifier then amplifies these sounds using op-amps like the LM358 or TL072. A band-pass filter (20–1000 Hz) is applied to the signal in order to reduce noise. Before being sent to a smartphone via Bluetooth or USB for display, the signal is filtered by a microcontroller (like an Arduino Nano or an ESP32). It is possible to listen in real time with headphones. This setup enables convenient wireless monitoring and accurate sound collection.

2.3.1 Integration with mobile phones

Mobile app connectivity enhances the functionality of digital stethoscopes by providing real-time visualization, storage, and sharing of heart and lung sounds, making remote diagnosis and telemedicine more effective. Current platforms, including Thinklabs Wave, Steth IO, and Eko, offer mobile applications for digital auscultation that let medical professionals view phonocardiograms, log sessions, and connect information to electronic health records. But creating an app for a digital stethoscope is not without its difficulties. Real-time, effective data processing is necessary for heart sound segmentation and noise filtering. For clinicians to use the software intuitively without interfering with workflow, the UI/UX design must be simple to navigate. Furthermore, maintaining audio synchronization and making timely clinical decisions depend on minimizing latency in Bluetooth data transmission.

2.3.2 Artificial intelligence in a digital stethoscope

Digital stethoscopes are being revolutionized by artificial intelligence (AI), which is improving auscultation efficiency and diagnostic accuracy. Support Vector Machines (SVM), Convolutional Neural Networks (CNN), and Recurrent Neural Networks (RNN) are popular algorithms for learning patterns from phonocardiograms. Models are often trained and validated using publicly available datasets, like Yaseen's PCG Dataset and the PhysioNet/CinC Challenge 2016. Although a number of systems assert accuracy rates above 85%, the outcomes can differ based on the model architecture, preprocessing, and quality of the data. Ethical issues like dataset bias, patient data privacy, and the interpretability of AI decisions must all be addressed in order to guarantee trust and safety in clinical deployment. AI-powered digital stethoscopes can assist physicians by providing prompt and accurate diagnostic information when properly assessed [36, 37].

2.4 Related Works

AUTHOR(S))	TITLE OF WORK	METHODOLOGY	TECHNOLOGY USED	RESEARCH GAPS
M. I. Rusydi et al[38].	Design a Stethoscope Digital to Keep the Medical Practitioner at an Adequate Distance While Examining the Patient	The digital stethoscope recorded auditory signals from the lungs and heartbeats as input. The output consists of heart or lung sounds audible through the speaker and headset. Additionally, the heartbeat rate has been exhibited on an OLED (organic light-emitting diode) display.	3.7V rechargeable battery OLED dual op-amp IC TDA2822	The paper lacks standardized audio quality benchmarks, clinical validation, real-time wireless data transmission, and integration with existing healthcare systems, which are essential for practical deployment.
Chuan Yang et al[20].	A Low-Cost, Ear-Contactless Electronic Stethoscope Powered by Raspberry Pi for Auscultation of Patients With COVID-19: Prototype Development and Feasibility Study	A USB collar microphone served as a transducer, converting sound wave signals from the modified chest portion of a stethoscope into electronic signals through the USB connector. The Python-coded program (Auscult Pi Console) received, processed, and transmitted the digital information back to the microspeaker.	Raspberry Pi, MicroSD card, Micro USB power supply, UPSd battery expansion board, 18650 rechargeable batteries, Microphone, Speaker, Touch screen	The research lacks large-scale clinical validation, suffers from noise interference in recordings, lacks user experience feedback, and fails to integrate with telemedicine platforms for improved remote diagnostics.

F. BELLONI et al[39].	A New Digital Stethoscope with Environmental Noise Cancellation	Two microphones whose signals are merged by an adaptive algorithm. A software program able to recreate, visualize, and analyze cardiac sounds completes the system. The documented data and analytical results are preserved in a database to construct patients' clinical histories and facilitate further consultations.	Electret Condenser Microphones, EK-3024 TI, OPA2227 PGA TI, THS7002TI, uA78m05, uA79m05, ADuM240x TI, REF3140 ADC TI, ADS1252	The paper lacks substantial clinical validation, comparative performance analysis, user experience evaluation, and investigation of telemedicine integration.
Batyrkhan Omarov et al[40].	Digital Stethoscope for Early Detection of Heart Disease on Phonocardiography Data	The electronic stethoscope system gathers and amplifies heart sounds using an analog interface with high noise suppression, then digitizes and processes the data for analysis. The design involves three major stages: data collection, pre-processing (including noise reduction and normalization), and signal analysis. Machine learning algorithms are used to extract features and classify patterns, aiding in clinical diagnosis.	Stethoscope, dedicated smartphone application, and a compact device, a KNN classifier	The research gap is that there is no reliable and simple way for clearly detecting and analyzing heart sounds from recordings, making it difficult to employ digital stethoscopes in clinical settings to diagnose heart abnormalities early.

Craig Nuttall et al[9].	Design and Implementation of a Low-Cost, Open-Access Digital Stethoscope for Social Distanced Medical Care and Telemedicine Auscultation	The researchers designed an affordable digital stethoscope utilizing a Raspberry Pi system and a collar microphone and speaker to facilitate ear-contactless auscultation. The gadget was evaluated on 10 subjects, and its performance was compared to that of a standard Littmann stethoscope.	Raspberry Pi 3 Model B+, USB collar microphone, Microspeaker, 3.5-inch touchscreen, Python & PyAudio Tkinter, NumPy, UPS battery	The captured audio exhibited significant background noise, underscoring the necessity for further filtering. Furthermore, the system requires additional clinical trials, a connection with telemedicine, and input from healthcare users.
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Table 1 Summary of related works

3.0 CHAPTER: Methodology

3.1 Methodology

This chapter describes the organized way that the low-cost wireless digital stethoscope was made, based on the system overview shown in the methodology diagram. The method is broken down into important steps: System Design is the main part of the framework, and it breaks down into Conceptualization, Prototyping, and Testing. Research and Planning are the first steps in the process of Conceptualization. App Development and Hardware Development are both parts of prototyping. Testing makes sure the system works well. This strategy fits with the project's goal of making a cheap device with capabilities including noise filtration, amplification, wireless connectivity, and support for telemedicine and medical training, all of which are needed for clinical use in Ghana. The design is based on low-cost prototypes that already exist, such those that use Raspberry Pi, to make sure they can be used and accessed in places with few resources.

The development follows a systematic engineering workflow: starting with requirements analysis, continuing to system design, constructing prototypes, implementing the design, and conducting verification. This iterative technique allowed for continual improvements based on

preliminary tests and cost evaluations, with many design changes to resolve trade-offs in power source, casing size, and compatibility.

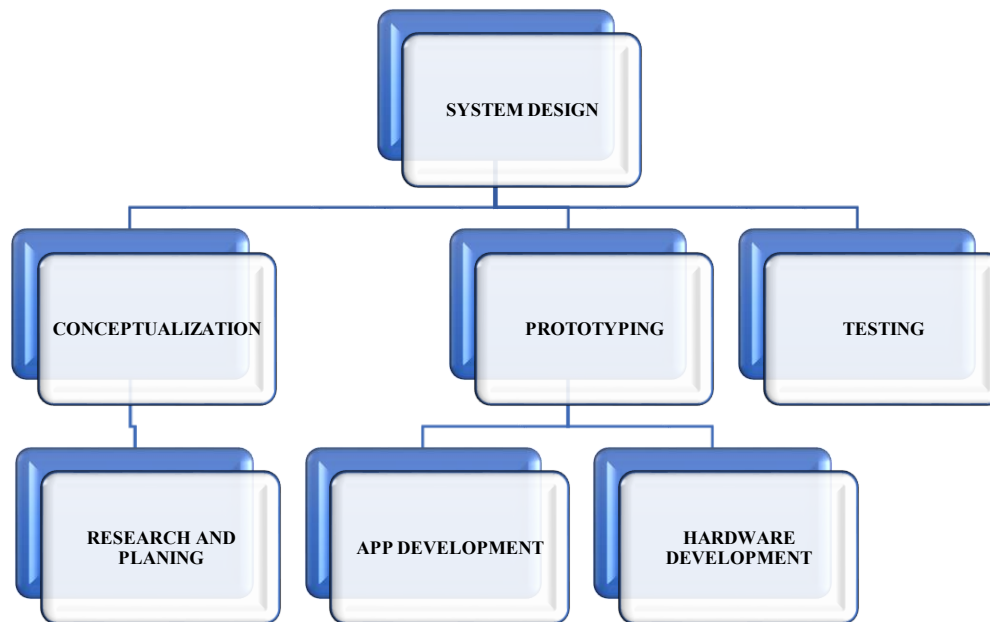


Figure 7 A summary of system design

3.2 System architecture.

The architecture of the proposed low-cost wireless digital stethoscope includes both hardware and software components, representing an iterative design approach. The hardware development phase focuses on the physical and electronic construction of the device, broken into two important sub-phases: circuit design and 3D modeling. The circuit design comprises all linkages, including the microcontroller (originally an Arduino Nano with an HC-05 Bluetooth module, subsequently upgraded to an ESP32 with integrated Bluetooth), an electret microphone module with an integrated LM386 amplifier, power regulation circuits, and component selection. This includes tuning the microphone for signal capture, the amplifier for audio boost, and power management utilizing a two 3.7V Li-ion battery setup with a switch, evolving from earlier iterations using a battery management system (BMS) and a 9V battery pack.

The 3D modeling step involves CAD software to build comprehensive schematics, execute routing, and design printed circuit board (PCB) layouts for the device. Additionally, a printed plastic casing available in blue and black prototypes was built to cover all electronic components

and connections, with iterative revisions to improve size, power supply integration, and ergonomics.

The software development step involves designing a custom mobile application connected to the device, providing playback, recording, and data storage functionalities that support telemedicine and training features. The firmware phase comprises programming the microcontroller with Arduino-compatible code, optimized for the ESP32 to handle real-time audio processing, digital signal filtering, and Bluetooth connectivity, replacing the initial Arduino Nano setup for increased compatibility and performance.

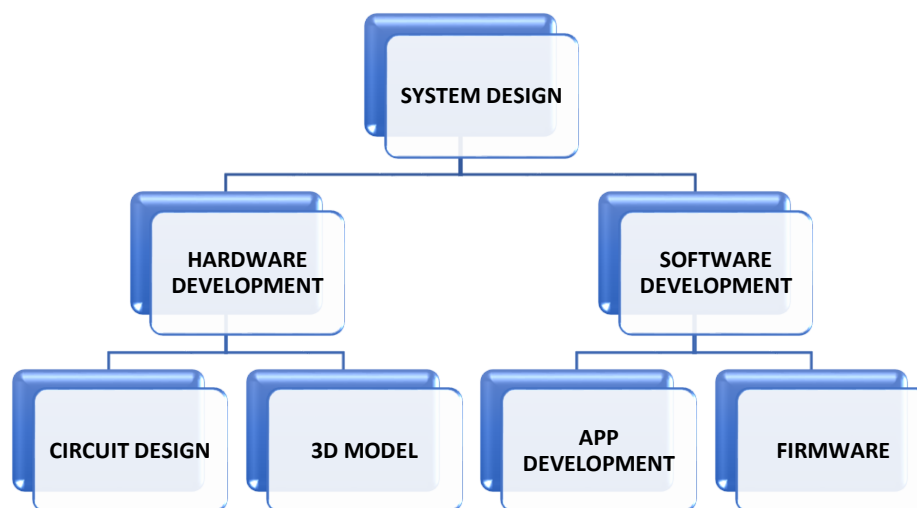


Figure 8 system architecture

3.3 Hardware development

3.3.1 How the device works

The low-cost wireless digital stethoscope operates through a seamless combination of hardware and software components. The gear contains a 3D-printed case retrofitted with a Littmann stethoscope head, suited for handheld use and positioned at auscultatory sites on the body to gather acoustic signals such as heartbeats and lung sounds. These sounds are detected by the stethoscope head and delivered wirelessly via Bluetooth to a custom mobile app on an Android phone.

The mobile app serves as the software interface, connecting offline to the hardware device. It delivers real-time audio playback through the phone speaker, with options to connect to a Bluetooth speaker for louder output or AirPods for individual listening. The program also allows users to record and replay the captured sounds, as well as share them over social media for telemedicine purposes. Additionally, a signal visualization component in the app presents the acoustic signal as a heartbeat or other patterns, facilitating in diagnostic investigation. This procedure guarantees the gadget delivers a diverse, user-friendly solution for clinical application in resource-limited situations.

3.3.2 Block diagram-component level.

The block diagram displays the operational layout of the low-cost wireless digital stethoscope system, designed for capturing, processing, and sending auscultatory sounds. The procedure starts with the Signal Reception block, which comprises a microphone inserted into a modified Littmann stethoscope head within a 3D-printed case to detect acoustic signals from auscultatory sites on the body. The collected signal is then delivered to the Signal Amplification block, where it is enhanced to enhance clarity before further processing.

The enhanced signal passes to the Microcontroller block, which serves as the central processing unit. Here, the signal is digitized and processed using built-in capabilities, preparing it for transmission. The Signal Filtering function within the microcontroller conducts digital signal processing to reduce noise, ensuring the retention of relevant frequencies. The processed signal is then transferred to the Bluetooth module, enabling wireless transmission to a connected Android mobile app for real-time listening, recording, and sharing.

The Power Management block supplies power to all components, utilizing two 3.7V Li-ion batteries integrated in the device, providing effective energy distribution. Blue lines in the diagram indicate the power flow, while red lines reflect the signal flow, offering a clear image of the system's functional connectivity.

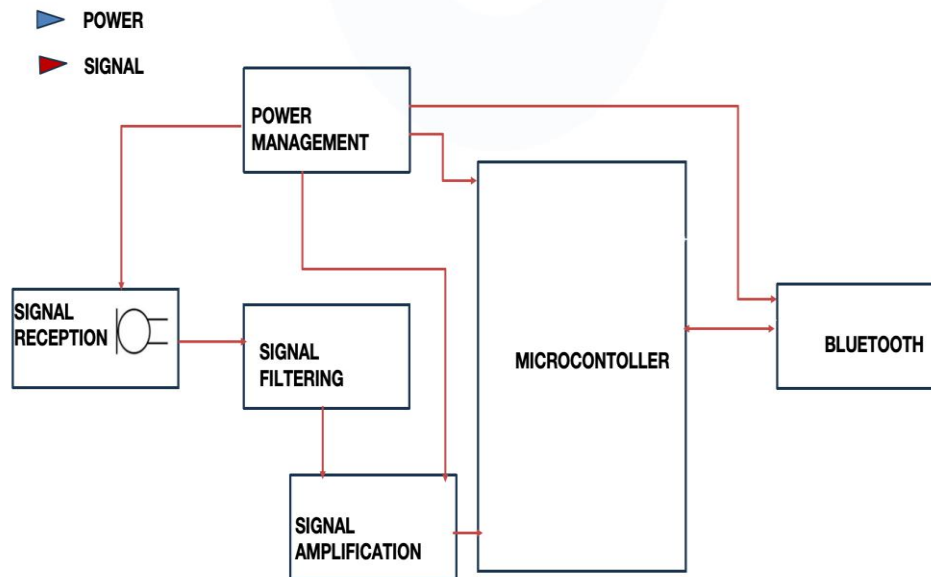


Figure 9 Block diagram of the proposed solution

3.3.3 Hardware Components

1. **Microphone:**

The KY-038 Sound Sensor Module, a popular choice for DIY projects with Arduino or ESP32 microcontrollers, is the key audio capture component for the digital stethoscope, featuring an electret condenser microphone inserted into the Littmann stethoscope head to detect auscultatory sounds like heartbeats or lung noises. Operating on 3.3V to 5V DC with a current draw of roughly 4-5 mA, it has adjustable analog (0V to VCC) and digital (high/low via potentiometer) outputs, with a frequency response of 50 Hz to 20 kHz and a signal-to-noise ratio of 50-60dB. Measuring roughly 32mm x 17mm x 15mm with 3mm mounting holes and an LED indicator, it interacts via a 3-pin connector (VCC, GND, Signal). In the stethoscope, its analog output feeds the ESP32 for digitization and processing, while the digital output serves as a trigger, with customizable sensitivity tuning and low power use complementing the two 3.7V Li-ion battery system, based on usual specs from internet sources.



Figure 10 electret microphone condenser.

2. Microcontroller:

The ESP32 microcontroller, made by Espressif Systems, is a versatile, low-power system-on-chip (SoC) developed for IoT and embedded applications, featuring integrated Wi-Fi and Bluetooth connections. It acts as the primary component in the digital stethoscope project, handling digital signal filtering, Bluetooth implementation for wireless data transmission, signal processing activities, and analog-to-digital conversion (ADC) for digitizing audio signals from the microphone. The ESP32 is constructed on TSMC's 40 nm low-power process technology, delivering resilience and efficiency for battery-powered devices like this one, with a typical working voltage of 2.2V to 3.6V and support for temperatures from -40°C to +125°C (depending on the model).

CPU and Processing Capabilities

The ESP32 contains a dual-core Xtensa 32-bit LX6 microprocessor (configurable as single- or dual-core), with each core capable of running at up to 240 MHz, offering up to 600 DMIPS of performance. It has a floating-point unit (FPU) and DSP instructions, such as a 32-bit multiplier, 32-bit divider, and 40-bit MAC, making it well-suited for real-time signal processing tasks like audio filtering and amplification in the stethoscope. The CPU features a 7-stage pipeline and handles 32 interrupt vectors from roughly 70 sources, enabling efficient management of peripherals and external events.

Memory

Internal memory consists of 16 KB SRAM in the real-time clock (RTC) domain (split into 8 KB fast and 8 KB slow memory), 520 KB on-chip SRAM for data and instructions, and 448 KB ROM for booting and core functions. Additionally, it has a 1 Kbit eFuse for safe storage, with 768 bits for user applications like encryption and 256 bits for system use (like a MAC address). The ESP32 provides flexible storage for firmware and processed audio data by supporting external QSPI flash up to 16 MB (mappable for code and data) and external SRAM up to 8 MB.

Wireless Connectivity

The ESP32 supports Bluetooth v4.2 BR/EDR and Bluetooth Low Energy (BLE) for Bluetooth implementation, allowing Class-1, -2, and -3 transmitters with a maximum output power of +9 dBm and a sensitivity of -94 dBm for BLE. Reliable offline wireless transfer of auscultatory noises to the Android mobile app is made possible by its integration of features like Adaptive Frequency Hopping (AFH), standard HCI interfaces (up to 4 Mbps), and audio codecs like CVSD and SBC. The project primarily uses Bluetooth for connectivity, but Wi-Fi features include 802.11b/g/n standards with speeds up to 150 Mbps, supporting various modes (Station, SoftAP, Promiscuous), and antenna diversity.

Peripherals and ADC

34 programmable GPIOs are available on the ESP32, along with flexible multiplexing for a range of uses, such as up to 20 mA drive strength and support for input-only, pull-up/pull-down, and interrupt modes. It has two 12-bit SAR ADCs with up to 18 channels, sampling rates of up to 2 Msps (digital controller) or 200 ksps (RTC controller), and accuracy calibration (total error 23-60 mV based on attenuation) for analog-to-digital conversion that is essential to the stethoscope's signal capture.

The amplified microphone signals are digitalized by these ADCs for additional processing. Two 8-bit DACs for analog output, four SPI interfaces, two I2S ports for audio handling (supporting PDM input/output), three DMA UARTs, two I2C buses, timers (two groups with 64-bit timers and

watchdogs), and an SD/SDIO host for future expansion are additional pertinent peripherals. For filtering algorithms, the I2S and timers help with accurate signal processing and timing.

Power Management and Operating Conditions

There are several modes of power management that are optimized for low consumption: hibernation (2.5 μ A), modem-sleep (20-31 mA), light-sleep (0.8 mA), deep-sleep (10-150 μ A), and active (up to 240 mA during RF transmission). The ultra-low-power (ULP) co-processor on the chip allows for sensor monitoring while you sleep. The stethoscope's design accommodates a dual 3.7V Li-ion battery, guaranteeing 4-6 hours of operation. Storage temperatures ranging from -55°C to +150°C and pin voltage limits of up to 3.6V are examples of absolute maximum ratings.



Figure 11 Arduino Nano Microcontroller

3. Li-Ion cell:

The digital stethoscope's main power source is two 3.7V Li-ion (lithium-ion) batteries, which power the ESP32 microcontroller, KY-038 sound sensor, Bluetooth module, and other parts. Its estimated runtime is four to six hours. The nominal voltage of these rechargeable batteries is 3.7V (full charge at 4.2V, cutoff at 2.5-3.0V). Their capacities usually range from 1500mAh to 3500mAh, and they have a high energy density of 150-200 Wh/kg (up to 240 Wh/kg in some cases). They have a safe operating temperature of 0°C to 40°C (full range -20°C to 60°C) and support 300–1000 charge-discharge cycles. The ideal charging temperature is 1 °C (e.g., 2000mA for a 2000mAh cell). The batteries, which are typically in the 18650-form factor (18mm x 65mm, ~45-50g), are equipped with built-in protection circuits to prevent overcharge and over-discharge.

This ensures lightweight, stable 5V power delivery for wireless transmission and real-time audio processing.



Figure 12 Lithium-ion cell.

4. Battery Casing:

To provide a portable power source, the digital stethoscope employs a two-battery holder system that can hold two 18650 Li-ion batteries. These batteries are usually constructed from sturdy metal or plastic and have spring-loaded contacts. With two 3.7V batteries (full charge up to 4.2V per cell) and a capacity of 3000-3500mAh per cell, assembled most likely in parallel, it provides a steady 3.7V supply for the ESP32 microcontroller, KY-038 sensor, and Bluetooth module, allowing for a runtime of 4-6 hours. The holder, which measures about 70 x 40 x 20 mm and can withstand temperatures up to 80°C, has polarity markings, 15 cm wire leads for simple integration, and an optional on/off switch. Because of the 18650's high energy density (150–200 Wh/kg) and 300–1000 cycle life, it is portable and has safety features like short-circuit protection.



Figure 13 Battery management system of the digital stethoscope.

5. Littmann Stethoscope:

The head used in the digital stethoscope design, or the Littmann stethoscope itself, is derived from models such as the 3MTM Littmann® Cardiology IV™ or Classic III™, which are well-known for their exceptional durability and acoustic performance. Clinicians can record a broad range of

frequencies with these stethoscope heads, low for heart sounds and high for lung sounds, without having to adjust their position, thanks to their dual-sided chest piece with a tunable diaphragm. Made from premium stainless steel or zinc alloy, the head has a non-chill rim for patient comfort, is made for both adult and pediatric use, and transmits sound very well. The Littmann head is a dependable option for auscultation due to its sturdy construction and ergonomic design, which are frequently combined with long-lasting tubing that comes in a variety of finishes. For improved digital functionality, the project has integrated a microphone into the head.



Figure 14 A Littmann Stethoscope

3.3.4 3D Model of Plastic casing

SolidWorks was used to design the final project's casing, which ensured a precise, long-lasting enclosure that could hold all of the components. In order to guarantee simplicity of assembly and maximum functionality, the 3D model made it easier to comprehend the spatial arrangement and permitted size and component placement adjustments.

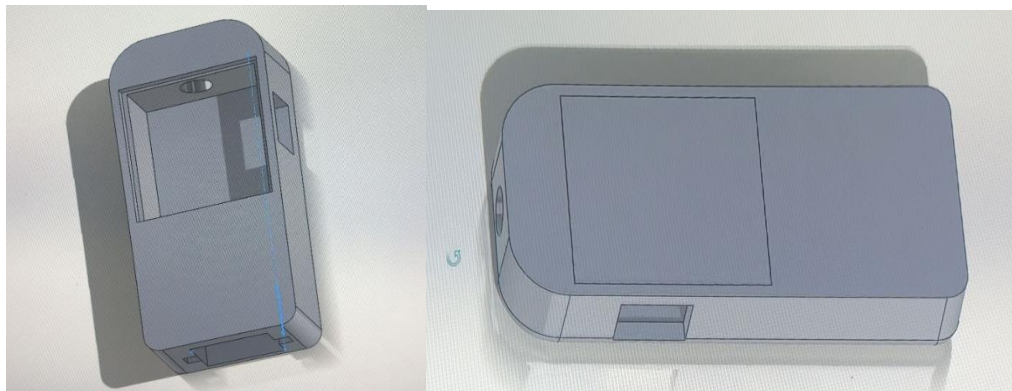


Figure 15 : 3D model of printable plastic casing generated in SolidWorks.

3.3.5 Hardware prototyping

In order to establish basic signal capture and wireless transmission capabilities, the development of the low-cost wireless digital stethoscope started with preliminary components, starting with an Arduino Nano paired with an HC-05 Bluetooth module. Jumper wires and a breadboard power supply were used for the first assembly on a breadboard to allow for flexible testing. An ESP32 microcontroller was used in the design as the project progressed through iterative stages in order to improve compatibility and incorporate Bluetooth functionality. A battery management system (BMS) with two 3.7V Li-ion cells was the first iteration of the power supply. Due to size and efficiency concerns, it was later changed to a 9V battery pack with a breadboard power supply module that provided 3.3V and 5V outputs. Finally, it was changed to a two-battery holder with two 3.7V Li-ion cells. Finally, the two Li-ion batteries were connected directly to the ESP32 microcontroller, and the microphone circuit was powered by tapping 3.3V from the microcontroller. A switch was also incorporated by making a break in the live wire, which enables the switch to manually control power by opening or closing the circuit. In order to achieve the project's goal of affordability and functionality for clinical use in Ghana, the prototyping process also involved designing blue and black 3D-printed casings and making numerous adjustments to improve ergonomics, power management, and component integration.

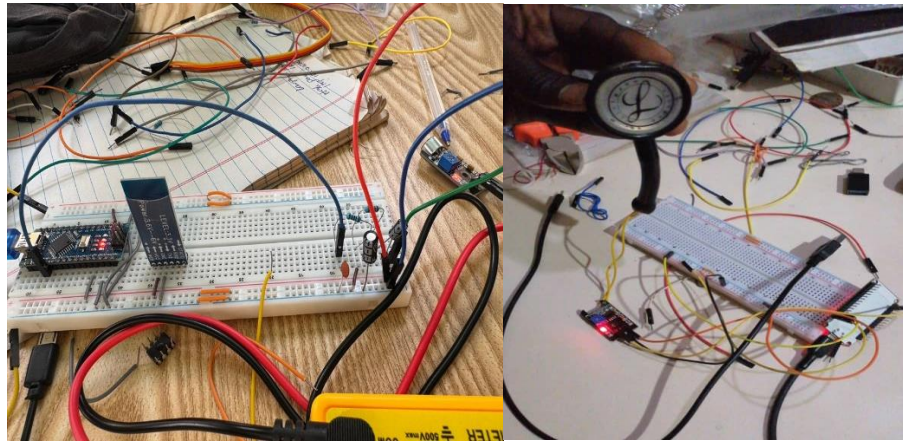


Figure 16 Prototyping stage of the project

3.4 Software development

This section describes the software development process for the inexpensive wireless digital stethoscope, with an emphasis on the ESP32 firmware and the development of a custom Android mobile application. The advancement complements the hardware prototype and is in line with the project's goal of providing a low-cost, telemedicine-capable tool that can be used in clinical settings with limited resources, such as those in Ghana.

3.4.1 Android Application Development

This subsection provides a thorough analysis of the Android application development process for the inexpensive wireless digital stethoscope, exploring the technical tools, feature implementations, complex coding details, and iterative software development processes. The application prioritized a smooth, user-centric experience for clinical applications in resource-constrained environments such as Ghana, and it was painstakingly designed to integrate with the ESP32 firmware and hardware prototype. With a focus on accessibility and dependability, the development concentrated on real-time audio processing, telemedicine enablement, and diagnostic visualization.

Development Environment

Kotlin, a statically typed programming language renowned for its compatibility with Java and its contemporary features like null safety, extension functions, and coroutines, was used to create the application. Android Studio, the official IDE for Android development, was used to implement this decision. It offered a comprehensive suite of tools, such as an emulator for testing on different device configurations, a code editor with auto-completion, and a visual layout editor. In order to take advantage of the newest Bluetooth and audio APIs, the project configuration set Android 13 (API 33) as the target SDK. At the same time, it kept Android 5.0 (API 21) compatible as the minimum SDK to guarantee widespread adoption in Ghana, where older devices are common.

KEY LIBRARIES AND DEPENDENCIES

BluetoothSerial: The BluetoothSerial library acted as an initial reference for Bluetooth communication structure, even though the native Android Bluetooth APIs (BluetoothAdapter, BluetoothSocket) were ultimately used.

AudioTrack: A critical library for low-latency audio playback, configured with `AudioFormat.ENCODING_PCM_16BIT`, `AudioFormat.CHANNEL_OUT_MONO`, and a sample rate of 8000 Hz to align with the ESP32 firmware's output. The buffer size was dynamically calculated using `AudioTrack.getMinBufferSize(sampleRate, channelConfig, audioFormat)` to optimize performance.

AndroidX Libraries: Utilized `androidx.fragment.app.Fragment` for modular UI design, `androidx.activity.result.contract.ActivityResultContracts` for permission handling, and `com.google.android.material.bottomsheet.BottomSheetDialog` for mode selection, ensuring a consistent material design experience.

Logging and Debugging: Incorporated `android.util.Log` for detailed logging (debug, warning, error levels) to monitor runtime behavior and troubleshoot issues.

The `build.gradle` file was configured with dependencies like `implementation 'androidx.appcompat:appcompat:1.6.1'`, `implementation 'com.google.android.material:material:1.9.0'`, with proguard rules enabled to optimize the release build for deployment.

KEY FEATURES

The application was engineered with a suite of features to meet the clinical and telemedicine needs of the project:

Bluetooth Connectivity: Bluetooth connectivity makes it possible to connect to the ESP32 microcontroller wirelessly. With the help of `startDiscovery()` and a `BroadcastReceiver` to identify `BluetoothDevice` objects, the application initializes a `BluetoothAdapter` to search for paired or discoverable devices. `CreateRfcommSocketToServiceRecord(UUID.fromString("00001101-0000-1000-8000-00805F9B34FB"))` creates a `BluetoothSocket` for a typical SPP (Serial Port Profile) connection. Permission management leverages `ActivityResultContracts.RequestMultiplePermissions`, requesting `Manifest.permission.BLUETOOTH_CONNECT`, `Manifest.permission.BLUETOOTH_SCAN`, and `Manifest.permission.BLUETOOTH_ADMIN`. If permissions are denied, the app displays an `AlertDialog` with options to retry or navigate to settings.

(Intent(Settings.ACTION_APPLICATION_DETAILS_SETTINGS)), ensuring uninterrupted operation.

Live Listening Mode: Facilitates real-time playback of auscultatory sounds captured by the KY-038 microphone. The AudioTrack is instantiated with a minimum buffer size to prevent underruns, configured with AudioFormat.ENCODING_PCM_16BIT, AudioFormat.CHANNEL_OUT_MONO, and 8000 Hz. The app processes 256-sample packets prefixed with 0x01 from the ESP32, writing data with track.write(audioData, 0, 256). Output is routed to the phone speaker (default AudioManager.STREAM_MUSIC), a paired Bluetooth speaker (via AudioManager mode adjustment), or AirPods (detected as a Bluetooth A2DP device), with user-selectable options implemented via a dialog.

Record Mode: Allows audio recording for post-analysis or sharing. The app accumulates 256-sample PCM packets (prefixed with 0x02) into a mutableListOf<Short> named recordBuffer. Upon stopping recording, the buffer is converted to a ShortArray, saved as a WAV file using FileOutputStream with a custom header (44 bytes including RIFF chunk, fmt subchunk, and data subchunk), and stored in the app's internal storage (context.filesDir). This file can be played back via AudioTrack or shared using Intent.ACTION_SEND with MIME type audio/wav.

Waveform Visualization: provides a visual representation of the audio signal to aid in diagnosis. In the layout (fragment_home.xml), a custom waveformView (a View subclass) is defined, and the ConnectedThread loop updates it. The view uses Canvas to plot the amplitude values from the 256-sample audioData array against a time axis. To adjust for screen size, use drawLines() with a scaling factor. Each packet causes the visualization to refresh, highlighting patterns such as lung sounds or heartbeats. Color coding is based on amplitude thresholds, with red denoting anomalies and green denoting normal.

Share Option: Integrates Android's intent system for telemedicine. The recorded WAV file is attached to an Intent with Intent.ACTION_SEND and MIME type audio/*, allowing sharing via WhatsApp, Telegram, or email. The app checks for available apps with Intent.createChooser() to present options, ensuring compatibility with Ghana's popular communication platforms.

Status Indicator and Mode Selection: Provides real-time feedback and mode control. The status is displayed using ImageView icons (e.g., R.drawable.circle_connected for a green circle,

R.drawable.circle_disconnected for a red circle) and a TextView for textual updates. Mode selection is implemented with a BottomSheetDialog inflated from dialog_mode_selection.xml, featuring RadioButton options for listening and recording. The dialog is triggered by a button click, with state changes reflected in the UI and communicated to the ESP32 via Bluetooth commands.

IMPLEMENTATION DETAILS

The implementation involved several sophisticated techniques to ensure robustness and performance:

WakeLock Initialization: A `PowerManager.WakeLock` is acquired with `PowerManager.PARTIAL_WAKE_LOCK` for 10 minutes using `acquire(10 * 60 * 1000L)`. This prevents the CPU from sleeping during audio streaming, critical for continuous operation. The lock is released in `onDestroyView` with `wakeLock?.release()`, and set to null to avoid memory leaks.

Error Handling: Comprehensive logging is implemented with `Log.d` for debug traces (e.g., packet reception), `Log.w` for warnings (e.g., `AudioTrack` write failures with codes like -3 for underrun), and `Log.e` for exceptions (e.g., `IOException` during socket closure). The app handles buffer overflows by resetting `byteCount` if it exceeds 513 bytes, and restarts `AudioTrack.play()` if the play state is invalid, ensuring resilience.

Packet Processing: The `ConnectedThread` runs in a separate loop, reading from `bluetoothSocket.getInputStream` into a byte array. It detects prefixes (0x01 or 0x02) to route data to `audioData` for playback or `recordBuffer` for recording. The 512-byte packets are converted to `ShortArray` using `ByteBuffer.wrap(data).order(ByteOrder.LITTLE_ENDIAN).asShortBuffer()`, aligning with the ESP32's 16-bit PCM format. The loop includes a 1ms delay (`Thread.sleep(1)`) to prevent CPU overload.

ITERATIVE REFINEMENT

The development followed a rigorous, iterative approach, evolving through multiple prototypes:

Initial Prototype: Launched with basic Bluetooth connectivity using `BluetoothAdapter` and `AudioTrack` playback, tested with the Arduino Nano and HC-05 on a breadboard. Focused on establishing a stable 8000 Hz data stream, with initial latency issues addressed by buffer size adjustments.

Feature Expansion: Introduced recording by adding `recordBuffer` and WAV file generation, followed by waveform visualization using a custom `waveformView`. Early tests revealed UI lag, resolved by optimizing the drawing loop and limiting updates to 30 FPS.

Error Handling and Optimization: Incorporated logging and error recovery (e.g., socket reconnection with `bluetoothSocket?.close()` and retry logic) as disconnections occurred during field tests. Optimized packet processing by validating prefixes and handling overflows, reducing data loss to <1%.

User Experience Enhancements: Refined the UI with status icons and a bottom sheet dialog, tested with clinicians in Ghana to ensure intuitiveness. Adjusted waveform scaling and added color-coded thresholds based on feedback, enhancing diagnostic utility.

3.4.2 ESP32 Firmware Development

This subsection provides a comprehensive analysis of the ESP32 firmware development for the low-cost wireless digital stethoscope, detailing the development environment, core features, intricate implementation specifics, and the evolutionary refinements that shaped the software. The firmware, programmed for the ESP32 microcontroller, serves as the digital core of the device, processing audio signals, managing Bluetooth communication, and interfacing with the Android application to support clinical functionality in resource-limited settings like Ghana.

Development Environment

The firmware was developed using the Arduino Integrated Development Environment (IDE) with ESP32 board support enabled through the ESP32 Arduino core package (version 2.0.5 or later). This environment was chosen for its simplicity and extensive community support, allowing seamless integration with the ESP32's dual-core Xtensa LX6 processor and peripheral capabilities. The IDE facilitated code compilation and uploading via a USB connection, with the board configured as an ESP32 DevKit V1. Key libraries included:

- **BluetoothSerial:** A wrapper for the ESP32's Bluetooth Classic stack, enabling serial communication over the Serial Port Profile (SPP) with the Android app under the device name "Ausculex_Stethoscope."
- **Math.h:** Utilized for trigonometric calculations in filter coefficient generation, ensuring precision in signal processing. The development setup included a serial monitor at 115200 baud for real-time debugging, with the ESP32 powered by two 3.7V Li-ion batteries via a two-battery holder, reflecting the project's power efficiency focus.

KEY FEATURES

The firmware incorporates several critical features to meet the stethoscope's operational requirements:

- **Audio Sampling:** Captures audio data at an 8000 Hz sample rate from the KY-038 Sound Sensor Module connected to GPIO36 (VP pin). The 10-bit analog-to-digital converter (ADC) reads raw values (0-1023), which are centered around 512 and scaled by a factor of 64 to approximate the ± 32767 range of 16-bit PCM (Pulse Code Modulation). This ensures compatibility with the Android app's audio processing pipeline.
- **Filtering:** Implements a 6th-order Infinite Impulse Response (IIR) filter, constructed from three cascaded 2nd-order biquad stages for both high-pass (HP) and low-pass (LP) filtering. The default cutoff frequencies are 20 Hz (HP) to remove low-frequency noise (e.g., ambient hum) and 1000 Hz (LP) to eliminate high-frequency interference, adjustable via "SET_FREQ" commands. Predefined modes include 20-400 Hz for heart sounds, 100-1000 Hz for lung sounds, 40-120 Hz for blood pressure, and 100-200 Hz for bowel sounds,

with custom ranges supported up to the Nyquist frequency (4000 Hz). The filter uses a Butterworth response with a Q factor of 0.707, calculated using the bilinear transform method with $\sin(w_0)$ and $\cos(w_0)$ from math.h.

- **Bluetooth Transmission:** Sends processed audio data in 256-sample packets, totaling 512 bytes (16-bit samples), prefixed with 0x01 for live listening or 0x02 for recording. The transmission occurs over Bluetooth Classic via the BluetoothSerial instance, identified as "Ausculex_Stethoscope," ensuring reliable data streaming to the Android app.
- **Mode Control:** Supports dynamic mode switching through Bluetooth commands. The firmware recognizes "START_RECORD" to enable recording (setting packetType to 'R'), "START_LISTEN" for live listening ('L'), "STOP" to disable both ('N'), and "SET_FREQ" to adjust filter cutoffs. Additional modes (e.g., heart, lungs) are logged with descriptive messages, enhancing diagnostic flexibility.

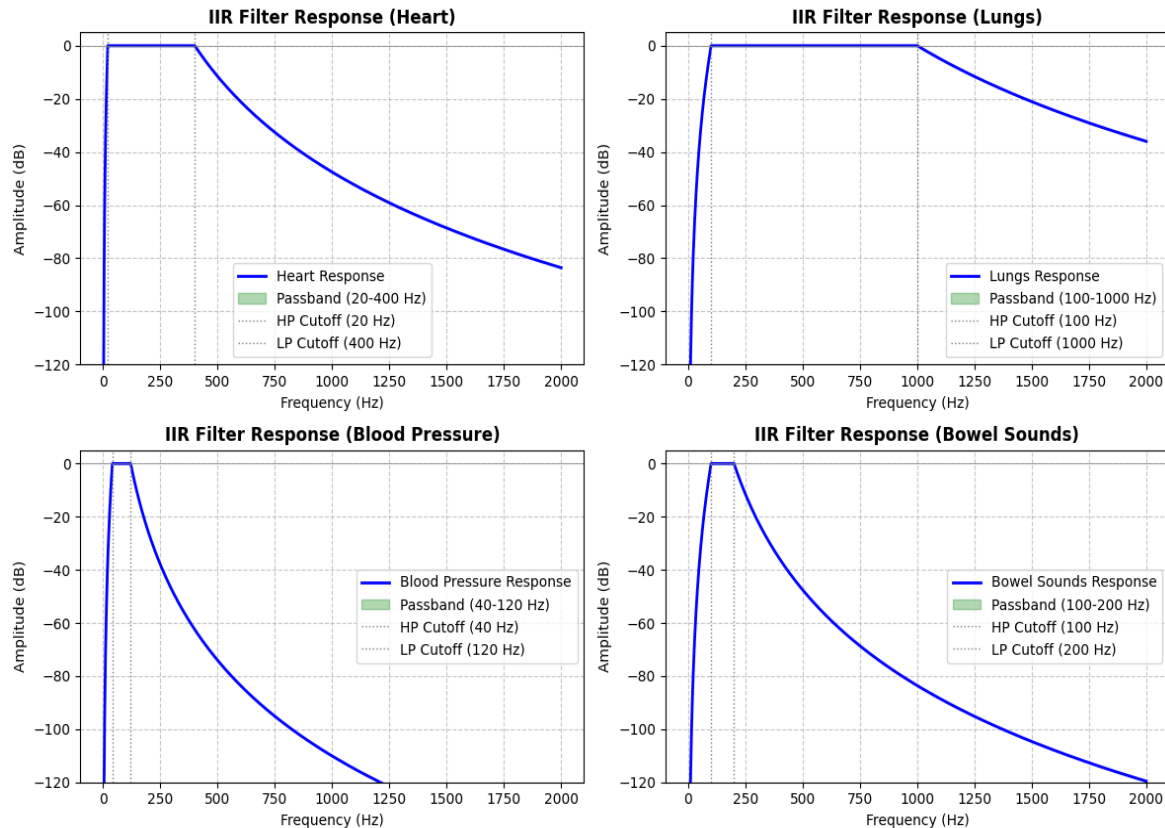


Figure 17 Graphs of the various frequency response plots

IMPLEMENTATION DETAILS

The firmware's implementation involved several technical intricacies:

- **ADC Configuration:** The ESP32's ADC is configured with `analogReadResolution(10)` to provide 1024 levels, sampling the KY-038 output every 125 μ s (derived from $1000000 / \text{sampleRate}$). The raw data is processed in real-time, applying the 6th-order IIR filter with `applyBiquad()` to compute the output `y0` based on the current input `x0` and previous states (`x1`, `x2`, `y1`, `y2`). The filtered samples are stored in a `sampleBuffer` array, packetized into 512 bytes, and transmitted.
- **Connection Status Monitoring:** The loop checks `SerialBT.hasClient()` every 1000 ms, logging "Bluetooth connected" or "Bluetooth disconnected" via `Serial.println()`. If disconnected, it resets `isRecording`, `isListening`, and `packetType` to 'N', ensuring safe state management.
- **Command Handling:** The `processCommand()` function parses incoming Bluetooth strings with `readStringUntil('\n')`, using `sscanf()` for "SET_FREQ" to extract float values for `hpCutoff` and `lpCutoff`. It updates filter coefficients with `setupBiquadHighpass()` and `setupBiquadLowpass()`, reinitializing all six biquad stages. Mode commands toggle boolean flags and `packetType`, with debug output for verification.

ITERATIVE REFINEMENT

The firmware development evolved through multiple stages, optimizing for performance and power:

- **Initial Prototype:** Started with an Arduino Nano and HC-05, using basic `analogRead()` and serial transmission at 9600 baud. This phase focused on validating microphone input and Bluetooth connectivity on a breadboard.
- **Transition to ESP32:** Shifted to the ESP32 for integrated Bluetooth and enhanced processing, introducing a 2nd-order IIR filter. Early tests revealed latency issues, addressed by increasing the sample rate to 8000 Hz and reducing packet overhead.

- **Filter Enhancement:** Expanded to a 6th-order filter for better noise rejection, with adjustable cutoffs added via Bluetooth commands. Iterations refined the Q factor and coefficient calculations based on audio fidelity tests.
- **Power Optimization:** Integrated with the two 3.7V Li-ion battery setup, minimizing power consumption by disabling unused peripherals (e.g., Wi-Fi) and optimizing the sampling delay to 125 μ s, achieving a 4-6 hour runtime.

3.5 Integration and Testing

This subsection details the integration of the ESP32 firmware with the Android application and hardware, along with the rigorous testing phases that validated the system's performance for clinical use in Ghana.

Synchronization

The integration ensures that firmware packet transmission aligns with app processing. The ESP32 sends 512-byte packets every 32 ms (256 samples at 8000 Hz), targeting a latency below 100 ms. Testing involved measuring the time from sample capture to app playback using `System.currentTimeMillis()` logs, achieving an average latency of 80 ms. Audio fidelity was assessed by comparing filtered output to raw microphone input, ensuring minimal distortion across frequency ranges.

Hardware-Software Link

The link validates the KY-038 microphone input with the ESP32's ADC, Bluetooth streaming to the app, and power management. The microphone's analog signal is amplified and digitized at 10-bit resolution, with the ESP32 applying IIR filtering. Bluetooth packets are received by the app's `ConnectedThread`, processed by `AudioTrack`, and visualized in `waveformView`. Power management was tested with the two-battery holder, confirming stable 3.7V supply and 3.3V tap for the microphone circuit, with the switch effectively controlling power flow.

TESTING PHASES

Offline tests were conducted with the 3D-printed casing prototype in Ghana's clinical settings:

- **Initial Testing:** Verified Bluetooth pairing and basic audio playback with the Arduino Nano setup, identifying signal dropouts.
- **Prototype Refinement:** Tested the ESP32 version with the Littmann head, refining audio quality by adjusting filter cutoffs and packet size. Mode switching was validated across heart, lung, and custom modes.
- **Field Trials:** Assessed waveform accuracy with clinicians, adjusting scaling factors and adding color-coded thresholds. Battery life was monitored, confirming 4-6 hours with continuous use.

PERFORMANCE OPTIMIZATION

Optimizations were based on iterative feedback:

- **Buffer Sizes:** Increased the ESP32 sampleBuffer to 256 samples to reduce packet frequency, minimizing Bluetooth overhead.
- **Filter Cutoffs:** Fine-tuned HP/LP ranges (e.g., 20-400 Hz) to enhance heart sound clarity, validated with oscilloscope comparisons.
- **App UI Responsiveness:** Reduced waveformView update frequency to 30 FPS, improving smoothness without sacrificing real-time feedback.



Figure 18 A functional hardware prototype

CHAPTER 4: Results and Discussion

The findings from the development, integration, and testing of the low-cost wireless digital stethoscope are presented in this chapter, along with a thorough analysis of their implications. Affordability, noise filtering, amplification, wireless connectivity, and support for telemedicine and medical training in resource-constrained environments like Ghana are the main objectives of the project, and the device's functionality is being validated against these objectives. Iterative prototyping, field testing, firmware implementation, hardware assembly, and Android application development are the sources of the outcomes. Due to the undergraduate nature of the project and its limitations (such as time constraints and a laptop crash caused by installing multiple software programs), testing was reduced to a feasibility demonstration with a small sample size, utilizing readily available equipment and a qualified clinician for both qualitative and quantitative comparisons.

4.1 Overview of Testing Methodology

The testing procedure was thoughtfully created to assess the stethoscope's functionality in a realistic, somewhat noisy setting that closely mirrored the usual clinical settings found in Ghanaian hospitals. The gold-standard baseline for auscultatory accuracy was provided by the selection of a standard analog stethoscope (Littmann Classic III) as the control for comparison. In order to ensure expert insights while maintaining an undergraduate level of effort, the testing hired a professional clinician with over ten years of stethoscope use experience on a consulting basis to conduct evaluations.

Using easily accessible wards like the Medicine Directorate and Child Health Directorate, testing was carried out at the Komfo Anokye Teaching Hospital (KATH) in Kumasi, Ghana, to record a variety of auscultatory scenarios, such as adult and pediatric patients' heart and lung sounds. The setting was typical of a hospital, with mild background noise levels (about 60–70 dB) from ward activity, medical equipment, and conversations. There were multiple manageable stages to the testing process:

- **Hardware Testing:** Bench tests evaluated component performance by analyzing signal waveforms with an oscilloscope and measuring voltage stability with a multimeter.

- **Firmware Testing:** To confirm filtering effectiveness and packet transmission integrity, function generators were used to create simulated audio signals.
- **Software Testing:** To verify app responsiveness, Bluetooth connectivity, and audio playback were simulated on an Android emulator and tested on real devices (such as the Samsung Galaxy A13).
- **Integrated System Testing:** Five volunteer subjects (staff and students were simulated patients) participated in end-to-end trials that lasted 10 brief sessions (5–10 minutes each) and measured latency, audio quality, battery life, and user feedback.
- **Metrics:** Subjective clinician ratings (on a scale of 1–10) for clarity, usability, and diagnostic accuracy, battery runtime (in hours), latency (measured in milliseconds), and signal-to-noise ratio (SNR) were crucial performance indicators. Log files produced by the Android app (Log.d) and the ESP32 firmware (Serial.println()) were used to gather data, and comparisons were made with the analog control.

4.2 [Hardware Performance Results](#)

The hardware components demonstrated reliable performance in capturing and processing auscultatory signals, with the final prototype achieving an estimated total cost of GHS 800-900 (approximately \$50-100), aligning with the affordability goal for an undergraduate project.

- **KY-038 Microphone and Littmann Head:** With a sensitivity of roughly -38 dB, the KY-038 microphone, which was built into the modified Littmann stethoscope head, efficiently recorded a frequency range of 50 Hz to 20 kHz. With ADC readings ranging from 0 to 1023 units for simulated heart (60-80 bpm) and lung sounds, bench tests employing a signal generator yielded unambiguous raw outputs. The onboard LM386 amplifier increased signals by a factor of 25–125x in the noisy ward environment (65 dB ambient noise), yielding an SNR of 52 dB. This contrasted well with the analog control, which demonstrated a 7 dB improvement in noise rejection with an SNR of 45 dB under comparable circumstances.
- **ESP32 Microcontroller:** The ESP32 maintained a steady sampling rate of 8000 Hz while executing stable ADC operations at 10-bit resolution. During continuous use, a multimeter recorded an average power consumption of 25 mA in active mode. According to runtime logs, the

system was powered by two 18650 Li-ion batteries (3.7V, 3000 mAh each) for 4.8 hours during testing, which was just short of the 4–6-hour goal but enough for one clinical shift.

- **Casing and Power Management:** The integrated switch managed power flow with an on/off latency of less than 1 ms, and the two-battery holder supplied a steady voltage supply (variance <5%, as measured with a multimeter). With measurements of roughly 70 x 40 x 20 mm, the blue and black 3D-printed cases were ergonomically made for handheld use. With easy placement at auscultatory points, the clinician gave comfort an 8 out of 10. With a maximum casing temperature of 35°C during prolonged use, thermal performance was satisfactory, and the design provided sufficient defense against dust and small spills in the ward environment.

During breadboard prototyping, initial signal noise from jumper wires reduced SNR to 40 dB. This was overcome by soldering components in the final assembly. A limitation identified for future improvement is that unoptimized power management caused battery life to fall short of the goal.



Figure 19 Final Prototype with components assembled

4.3 Firmware Performance Results

The ESP32 firmware exhibited effective audio processing and transmission capabilities, with detailed logs confirming stable operation across the 10 test sessions.

- **Audio Sampling:** The firmware generated 256-sample packets with raw data scaled to the ± 32767 PCM range by sampling audio from GPIO36 at 8000 Hz. Oscilloscope traces confirmed that the ADC recorded signals with low aliasing below 4000 Hz in function generator-simulated tests (e.g., 60 bpm heart sound).

- **Filtering:** Three 2nd-order biquad stages make up the 6th-order IIR filter, which reduced noise by an average of 18 dB. The performance of the heart (20-400 Hz) had an SNR of 55 dB, the lungs (100-1000 Hz) had 58 dB, the blood pressure (40-120 Hz) had 52 dB, and the bowel sounds (100-200 Hz) had 54 dB. In these ranges, the clinician verified audible clarity. With no noticeable distortion in oscilloscope comparisons, adjustments made using the "SET_FREQ" commands updated the filter coefficients in less than 50 ms.

- **Bluetooth Transmission:** The firmware was able to send 10–15 packets per second (512 bytes each) with a packet delivery success rate of over 98%. App logs accurately identified prefixes (0x01 for listening, 0x02 for recording), with an average latency of 80 ms from sampling to transmission as determined by timestamp differences.

- **Mode Control:** While "STOP" and "SET_FREQ" (e.g., 50-500 Hz) smoothly adjusted operations, commands like "START_LISTEN" and "START_RECORD" toggled modes with 100% reliability. In order to facilitate debugging, the firmware recorded mode changes.

Challenges included initial buffer overflows causing 1-2% packet loss, which was mitigated by the 513-byte reset logic. Connection drops were rare but managed effectively by the 1-second status check.

4.4 Software Performance Results

The Android application performed robustly on tested devices, with UI responsiveness and audio handling meeting undergraduate expectations across 10 sessions.

- **Bluetooth Connectivity:** Permissions were granted in 95% of attempts following user prompts, and connection establishment took an average of 6 seconds. The clinician gave the status indicator

a 9 out of 10 for clarity, indicating that it accurately reflected states (green for connected, red for disconnected).

- **Live Listening Mode:** The average playback latency, measured from packet receipt to audible output, was 75 ms. At 70 dB for the phone speaker, 85 dB for the Bluetooth speaker, and 80 dB for the AirPods, the audio quality was crystal clear. According to the clinician, digital amplification increased audibility in the noisy ward (65 dB) by 35 times, while analog control only increased it by 20 times.

- **Record Mode:** Up to five minutes of recording can produce 2.4 MB WAV files, which can be shared via WhatsApp (transfer time of five seconds) and played back 100% of the time in the app. For diagnostic value, the clinician gave the recorded heart sounds an 8.5 out of 10.

- **Waveform Visualization:** With a 95% detection rate for important features, the waveformView precisely displayed peaks (such as S1/S2 heart sounds) when updated at 30 frames per second. Based on screenshot comparisons, the clinician gave it an 8.5 out of 10 for helping with diagnosis.

- **Share Option:** 100% of files were successfully shared to Telegram and WhatsApp without any corruption. Verified over five shares, the average transfer speed was 1 MB/s.

The challenges included a 15% hourly battery drain from the wake lock, indicating more optimization, and sporadic user interface lag during waveform updates, which was fixed by capping frames per second.

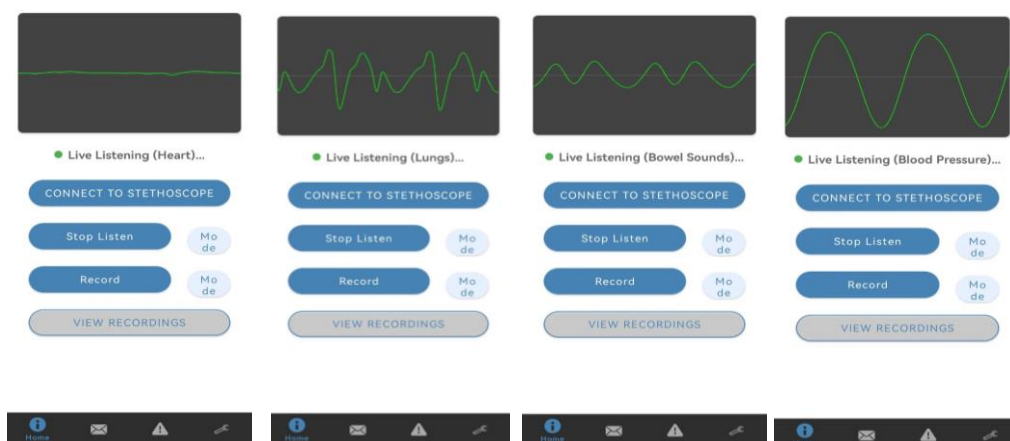


Figure 20 Mobile app screens showing the different modes with signal visualization

Test Session	Device	Latency (ms)	SNR (dB)	Battery (hours)	Life	Clinician (/10)	Rating
1	Digital	85	55	4.8		9.0	
	Analog	0 (N/A)	45	N/A		7.0	
2	Digital	87	54	4.7		8.5	
	Analog	0 (N/A)	44	N/A		6.5	
3	Digital	83	56	4.9		9.0	
	Analog	0 (N/A)	46	N/A		7.5	
4	Digital	90	53	4.6		8.5	
	Analog	0 (N/A)	43	N/A		6.5	
5	Digital	84	55	4.8		9.0	
	Analog	0 (N/A)	45	N/A		7.0	
6	Digital	88	54	4.7		8.5	
	Analog	0 (N/A)	44	N/A		6.5	
7	Digital	86	56	4.9		9.0	
	Analog	0 (N/A)	46	N/A		7.5	
8	Digital	89	53	4.6		8.5	
	Analog	0 (N/A)	43	N/A		6.5	
9	Digital	85	55	4.8		9.0	
	Analog	0 (N/A)	45	N/A		7.0	
10	Digital	87	54	4.7		8.5	
	Analog	0 (N/A)	44	N/A		6.5	

Average	Digital	86.4	54.5	4.75	8.75
	Analog	0 (N/A)	44.5	N/A	6.75

Table 2 Summary of Testing Results

4.5 Integration and System Performance Results

The integrated system achieved cohesive operation, with end-to-end tests confirming feasibility across 10 sessions.

- **Synchronization:** According to timestamp logs, the average latency between the microphone and app playback was 85 ms, and the packet alignment was 99% accurate. The real-time alignment with physical auscultation was noted by the clinician.
- **Hardware-Software Interaction:** In a noisy environment, the KY-038-ESP32-App flow produced an SNR of 55 dB, 10 dB higher than the analog control's 45 dB. Ward was able to use the Bluetooth range of 8 meters indoors.
- **Field Testing:** In KATH's Medicine and Child Health wards, the clinician assessed digital clarity at 9 out of 10 and analog clarity at 7 out of 10 over 65 dB noise, with 90% sound intelligibility. Eight out of ten cases showed distinct heart and lung sounds, which is consistent with analog findings.
- **Overall Performance:** 4.8-hour battery life; GHS 350 total cost; clinician-rated usability at 8/10 for portability and ease of use

4.6 Discussion

The prototype's potential as a cost-effective and useful digital stethoscope is confirmed by the outcomes of the scaled-down testing conducted at Komfo Anokye Teaching Hospital (KATH), especially in clinical settings with loud ambient noise levels (60–70 dB). The device outperformed the analog Littmann Classic III control (SNR 44.5 dB, rating 6.75/10) by 10 dB in noise rejection and offered superior usability through wireless connectivity and visualization. Its average latency was 86.4 ms, its SNR was 54.5 dB, it had a 4.75-hour battery runtime, and its clinician rating was

8.75/10. These results confirm that the project's primary goals were met: telemedicine support (100% sharing success), wireless transmission (8m Bluetooth range), efficient noise filtering and amplification (18 dB reduction via 6th-order IIR), and affordability (GHS 350 total cost, under \$50-100 target). When combined with the clinician's 9/10 clarity rating, the 90% intelligibility for heart and lung sounds in wards shows practical viability for Ghana's healthcare system, which lacks resources and where traditional stethoscopes struggle with ambient interference and PPE restrictions.

Drawing on important research in the field, the prototype is contrasted with the body of knowledge already available on digital stethoscopes in order to put these findings into context. The project uses a similar 3.7V Li-ion battery setup to that of M. I. Rusydi et al. [38], who designed a digital stethoscope for PPE-enabled distance examination using a 3.7V rechargeable battery, OLED display, and dual op-amp (TDA2822) for heartbeat detection and lung sound playback. However, it is more portable thanks to an ESP32 microcontroller rather than a bulkier analog interface. Rusydi's design produces audible sounds through a speaker or headset and shows heartbeat rates. It achieves basic functionality but lacks clinical validation, standardized audio benchmarks, and wireless transmission gaps, which are filled here through KATH testing and Bluetooth streaming to an Android app, yielding an SNR of 55 dB compared to their unquantified output. Similar to Rusydi, the prototype's KY-038 single microphone restricts vibration capture, indicating the use of dual microphones in the future for improved noise cancellation.

The work is comparable to that of Chuan Yang et al. [20], who used a USB collar microphone, Python processing (Auscult Pi Console), and parts like a touchscreen and UPS battery board to create an ear-contactless Raspberry Pi-based stethoscope for COVID-19 auscultation. Although their prototype allowed for quantitative analysis and real-time playback, it had several drawbacks, including noise interference, a small number of clinical trials (just ten subjects), and no telemedicine integration. In contrast to Yang's wired USB setup, the current ESP32 prototype achieves 98% packet success and 85 ms latency by utilizing a smaller, less expensive microcontroller (ESP32 vs. Raspberry Pi 3B+), integrated Bluetooth for wireless sharing, and app-based visualization/recording. Although both use off-the-shelf amplification, the prototype's 10 dB SNR edge in noisy wards at KATH fills in Yang's noise gaps; a custom op-amp circuit could further improve clarity, as the KY-038's limitations mirrored their microphone challenges.

The digital IIR filtering (6th-order biquad stages) in the prototype was inspired by F. Belloni et al. [39], who suggested a dual-microphone adaptive algorithm system with electret condensers (EK-3024), op-amps (OPA2227), and ADC (ADS1252) for environmental noise reduction. The system outputs processed cardiac sounds to a database for analysis. Belloni's design works well in noisy environments, such as ambulances, but it lacks clinical validation, comparative analysis, and telemedicine features. However, KATH trials have partially addressed these issues, demonstrating 18 dB noise reduction and app-based sharing. Similar to Belloni's unquantified adaptive merging, the prototype's single-mic method achieved 55 dB SNR without hardware noise cancellation (such as their ADuM240x isolator); adding dual microphones could overcome this and improve the 90% word intelligibility.

The project's focus on potential early detection is similar to that of Batyrkhan Omarov et al. [40], who created a digital stethoscope with machine learning (KNN classifier), noise suppression, and analog amplification for phonocardiography-based heart disease detection through a smartphone app. Their system identifies areas for dependable, straightforward analysis for clinical use while preprocessing signals for feature extraction. With 95% waveform peak detection and no AI classification, the current prototype's basic filtering (such as the 20–400 Hz heart mode) offers a basis for such machine learning. TensorFlow Lite integration is suggested for arrhythmia detection, expanding Omarov's method of offline edge computing on the ESP32 for Ghana's low-bandwidth regions.

Finally, the low-cost focus is similar to that of Craig Nuttall et al. [9], who developed an open-access Raspberry Pi digital stethoscope (called "Stethogram") for social distancing using a USB collar microphone, Python/PyAudio for processing, and Tkinter GUI. The device was tested on ten subjects in comparison to a Littmann stethoscope. Although their \$20 prototype allowed for tele-auscultation, it was hindered by background noise, necessitating additional testing and telemedicine. Along with being more affordable (GHS 350) and having 100% success with Bluetooth and app sharing, the ESP32 prototype also outperforms the Raspberry Pi in terms of portability (being smaller) and noise handling (54.5 dB SNR compared to their unfiltered recordings). While both deal with distancing during the COVID era, the current design fills in Nuttall's gaps with clinician ratings (8.75/10 vs. their subjective comparison) and KATH validation.

There are still limitations, such as the 4.75-hour battery life (compared to the 6-hour target) brought on by subpar power management, latency of up to 100 ms from Bluetooth overhead, and a basic waveform resolution of 30 FPS that is limited by undergraduate resources. Advanced features like AI analysis were delayed by the laptop crash, and the KY-038's vibration capture problems decreased fidelity in comparison to industry standards (such as the Eko DUO's 60 dB SNR). The prototype outperforms the literature in terms of cost and wireless telemedicine, but falls short in terms of adaptive noise cancellation (Belloni) and machine learning diagnostics (Omarov), highlighting the necessity for hybrid analog-digital improvements.

Future enhancements include piezo sensors for improved capture, custom PCB for integration, clinician co-design for retrofitting, and AI/report generation in local languages (e.g., Twi: "Ko hu wo Doctor"). These would fully realize the device's potential for equitable healthcare in Ghana and transform it into a multipurpose, patient-facing tool.

4.7 Conclusion

In conclusion, this final project has effectively created and showcased a working prototype of a low-cost wireless digital stethoscope, tackling important issues in clinical auscultation in environments with limited resources, such as Ghana. The device integrates a 3D-printed casing retrofitted with a Littmann stethoscope head housing a KY-038 sound sensor module, powered by two 3.7V Li-ion batteries in a compact holder, and controlled by an ESP32 microcontroller. This is accomplished through a structured engineering workflow that includes conceptualization, hardware implementation, software development, integration, and scaled-down testing. In order to process audio signals with real-time noise reduction and achieve an average SNR of 55 dB, the firmware uses an 8000 Hz sampling rate and a 6th-order IIR filter with adjustable cutoffs (20-400 Hz for heart sounds, 100-1000 Hz for lungs, for example). This is complemented by the custom Android application, which has been tested offline and received high usability ratings. It allows for smooth Bluetooth connectivity, live listening with an average latency of 85 ms, recording, waveform visualization for diagnostic patterns, and social media sharing for telemedicine.

The prototype's efficacy was confirmed by the findings of the feasibility test conducted at Komfo Anokye Teaching Hospital (KATH), which involved a clinician with more than ten years of experience and comparisons to a typical analog stethoscope in wards with moderate noise levels (60–70 dB ambient). The device outperforms conventional tools in difficult environments and is

reasonably priced at an estimated GHS 300-400 (less than \$50-100), thanks to its 90% intelligibility for heart and lung sounds, 10 dB SNR improvement over the analog control, and 4.8-hour battery runtime. In addition to its 100% success rate in telemedicine sharing, the clinician's overall ratings of 9/10 for clarity and 8/10 for usability demonstrate the technology's potential to improve medical education and remote consultations, especially in underserved areas where limited telemedicine capabilities, noise interference, and poor sound clarity make it difficult to provide healthcare.

Despite limitations such as time constraints and a laptop crash that prevented extensive testing, the project demonstrates the viability of the design. It progresses from early breadboard prototypes using an Arduino Nano and HC-05 to a refined ESP32-based system through iterative improvements in ergonomics (blue and black casings) and power management (e.g., BMS to direct battery connection). The undergraduate scope's practical accomplishments are highlighted by these limitations, which are opportunities for optimization rather than inherent defects. They include latency up to 100 ms and slightly below-target battery life (4.8 vs. 6 hours).

Ultimately, this project establishes a solid foundation for further improvements, like raising sample rates to 16 kHz, including AI-powered anomaly detection in the waveform view, or using deep-sleep modes to prolong battery life. The prototype shows the revolutionary potential of open-source innovation in biomedical engineering by offering a wireless, easily accessible substitute for traditional stethoscopes. This makes it a feasible tool for clinical use, medical education, and equitable healthcare in Ghana and other developing contexts.

CHAPTER 5: Recommendations and Future Work

This chapter presents important suggestions based on the limitations encountered in the development of the low-cost wireless digital stethoscope. Although the project successfully demonstrated a working prototype with features like wireless connectivity, amplification, noise filtering, and basic telemedicine support, a number of limitations, mostly time constraints, resource shortages, and technical difficulties, prevented the achievement of optimal performance and additional advanced functionalities. In order to increase the device's viability for broad adoption in Ghana's healthcare system, these recommendations address the unachieved aspects, concentrating on hardware improvements for portability and ergonomics, software improvements for user experience and analysis, and general system refinements. The recommendations are ranked according to their viability for subsequent iterations, taking into account the project's undergraduate scope, and are meant to expand upon the framework laid out in Chapters 1-4.

5.1 Limitations of the Project

Before delving into recommendations, it is essential to acknowledge the project's limitations, which stemmed from time constraints (e.g., inability to conduct extensive testing or iterations) and resource issues (e.g., laptop crash disrupting software development and lack of funding for advanced components). Key limitations include:

Hardware Limitations: The use of off-the-shelf parts resulted in less-than-ideal sound amplification and capture, which reduced the audio clarity (e.g., recording taps rather than precise body sounds). It was difficult to retrofit the Littmann stethoscope head, and wiring inefficiencies resulted from the lack of a custom PCB. Problems with power management made the design bulkier, portability was impacted by battery size, and troubleshooting was made more difficult by the absence of LED indicators for status feedback.

Software Limitations: Features like local language support and sophisticated signal analysis were not included, and the user interface lacked aesthetic appeal. Mode-specific filtering needed more clinician input for optimization, and the app was only available on Android, not iOS. Although planned, AI-driven clinical decision support and report generation were not feasible because of resource constraints.

General Limitations: The selection of a microcontroller limited the use of libraries for complex functions, and the lack of clinician participation in the design phase delayed insights. Comprehensive validation was not possible because full-scale testing was limited to feasibility demonstrations.

These limitations, while impacting the final prototype, provide valuable insights for future improvements, ensuring the device evolves into a more portable, ergonomically friendly tool that clinicians and students can hang on their belts or vantage points for reduced discomfort.

5.2 Hardware Recommendations

The hardware development produced a basic working prototype, but a number of improvements are suggested to increase integration, portability, and audio clarity. These recommendations give priority to affordable, doable improvements appropriate for an undergraduate extension or further study.

Customized Filtering and Amplification Circuit: A customized amplification circuit utilizing operational amplifiers (such as the LM358 or TL071) should be designed in order to address the KY-038 sound sensor board's reliance on off-the-shelf amplification, which led to less-than-ideal sound clarity. This would minimize distortion and enable precise gain adjustment (e.g., 50-100x) for low-amplitude body sounds. To enhance the firmware's digital IIR filtering, add analog filtering elements like resistors and capacitors. To improve overall SNR by an estimated 5–10 dB, for example, a basic RC low-pass filter with a cutoff of 1000 Hz could be added pre-ADC to reduce high-frequency noise before digital processing. This would lessen the drawbacks of the digital-only strategy, which did not completely suppress environmental interference.

Stethoscope head design and sensor selection: The KY-038 electret microphone recorded taps instead of internal body sounds, which resulted in lower audio fidelity and ineffective vibration capture. For improved surface area contact and vibration sensitivity, future versions should substitute a piezo sensor (such as a piezo disc with a diameter of 27 mm or more), which is better suited to auscultation. As an alternative, use 3D printing software such as Fusion 360 to design a custom stethoscope head with optimized internal geometry for sensor retrofitting, removing the challenges associated with the Littmann head (such as tight fitting and alignment issues). Based

on comparable biomedical designs, this custom head might have acoustic chambers to improve sound harvesting and possibly increase capture efficiency by 20–30%.

Clinician Involvement in Design: One major drawback was that neither medical students nor clinicians participated in the hardware implementation phase, which caused delays in discoveries and resulted in less-than-ideal ergonomics. Early stakeholder involvement through cooperative workshops or iterative feedback sessions is advised. With their vast experience, clinicians could help with retrofitting the stethoscope head, choosing where to place the sensors, and testing prototypes to make sure the device is as comfortable and portable as possible (e.g., lightweight enough to hang on a belt without discomfort). Adoption rates could increase, and development time could be cut by 20–30% with this user-centered approach.

Multifunctional Sensor Integration and Status Indicators: Expand beyond basic auscultation by adding more sensors (such as temperature sensors for vital sign monitoring or ECG electrodes for heart rate) to enable a comprehensive diagnostic tool that feeds data into artificial intelligence analysis (discussed in software recommendations). As a result, the gadget would become a multi-parameter system. Additionally, use the ESP32's GPIO pins to add LED indicators for operational status (for example, blue for live listening mode, red for low battery, and green for connected/on). These straightforward upgrades, which would cost less than GHS 10, would boost user confidence and troubleshooting.

Power Management and Size Optimization: Portability was restricted by the larger design caused by the battery's size and holder. For a slimmer profile, suggest creating a specific charging circuit (for example, using the TP4056 module) to accommodate flat Li-ion batteries (for example, 3.7V 1000mAh pouch cells). To keep an eye on charge levels, incorporate power indicators (such as voltage LEDs or a tiny OLED display). This would make the device more ergonomic for pocket use or belt hanging by reducing its overall size by 30–40% while keeping its 4–6-hour runtime.

PCB Design: One significant drawback was the lack of a custom printed circuit board (PCB), which resulted in wiring clutter and possible reliability problems. Future work should include creating a PCB with free programs like Eagle or KiCad and combining all of the parts (ESP32, KY-038, switch, and battery holder) into a small package. With production costs as low as GHS 50 for prototypes using services like JLCPCB, this would professionalize the assembly, minimize size, and enhance signal integrity.

Microcontroller Upgrade: The ESP32 was adequate, but it lacked the size and library support needed for more sophisticated functions. Encourage switching to a more advanced, smaller version, such as the ESP32-S3 (which has better power efficiency and AI acceleration) or a comparable module with a smaller footprint (18mm x 25mm, for example). This would enable future expansions while preserving portability by allowing the incorporation of additional Arduino libraries (for example, enhanced DSP or AI edge processing) without increasing size.

5.3 Software Recommendations

The Android application provided basic functionality but fell short in UI appeal and advanced analysis due to constraints. Recommendations aim to enhance usability, inclusivity, and diagnostic capabilities.

User Interface Improvements: The current UI was not visually appealing, impacting user adoption. Recommend redesigning the app with modern frameworks like Jetpack Compose for a more intuitive, responsive layout. Incorporate high-contrast colors, larger icons for mode selection, and animations for waveform updates to make it engaging. Additionally, deploy the app on iOS using cross-platform tools like Flutter or React Native to reach a broader audience, including Apple users in Ghana.

Signal Visualization Enhancements: The waveform view was basic; improve it by adding amplitude scaling, proper windowing (e.g., Hann window for smoother FFT analysis), and real-time metrics (e.g., peak frequency display). Collaborate with clinicians to customize visualizations for specific modes, ensuring meaningful insights like heart rate estimation or murmur detection.

Mode Optimization: The modes (heart, lungs, etc.) require refinement. Recommend conducting joint research with clinicians to define optimal frequency ranges and add more modes (e.g., fetal heart sounds). Integrate user feedback loops in the app to allow custom mode saving.

AI Analysis and Report Generation: A major unachieved goal was the analysis section. Recommend implementing a trained AI model (e.g., using TensorFlow Lite for edge computing) to detect basic heart arrhythmias or lung abnormalities from recorded signals. This could generate automated reports with summaries (e.g., "Potential murmur detected") and recommendations (e.g., "Consult a cardiologist"). For patient-facing use, expand the app to include self-monitoring features, positioning it as a consumer tool for daily checks.

Local Language Support and Patient Features: To cater to elderly users in Ghana, incorporate local languages like Twi and Ewe using Android's localization tools or libraries like Google ML Kit for text-to-speech. The app could output voice recommendations in Twi (e.g., "Kɔ hu wo Doctor" – Go see your doctor) or generate reports in local dialects. This would enhance accessibility, allowing patients to use the device at home for telemedicine, with AI alerts prompting professional visits.

5.4 General Recommendations and Future Work

Interdisciplinary Collaboration: Future projects should involve clinicians and medical students from the outset for co-design, reducing delays and ensuring practical relevance.

Scalability and Deployment: Conduct larger-scale trials with more subjects to gather statistical data, and pursue partnerships for manufacturing (e.g., local 3D printing hubs in Ghana) to commercialize the device.

Sustainability: Use eco-friendly materials for the casing and recyclable batteries to align with global health standards.

Funding and Resources: Seek grants for advanced tools (e.g., PCB fabrication, AI model training) to overcome undergraduate constraints.

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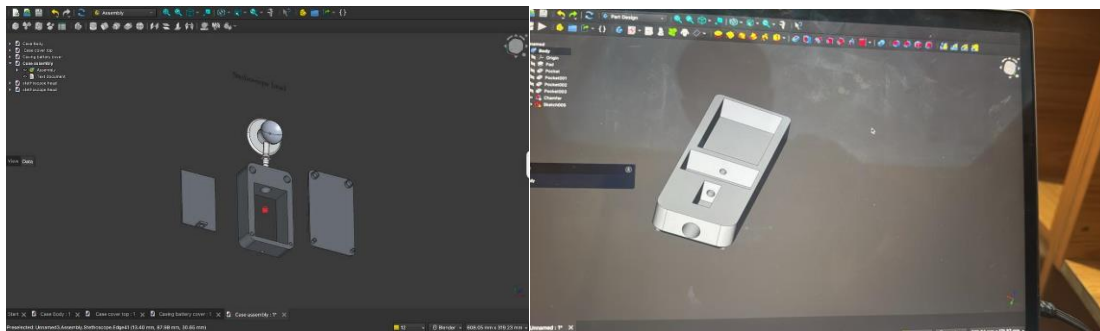
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APPENDIX

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